MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, LAMP/RELAY, DIGITAL, KEY CODE FQB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic lamp/relay driver module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.986 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FQB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/206-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

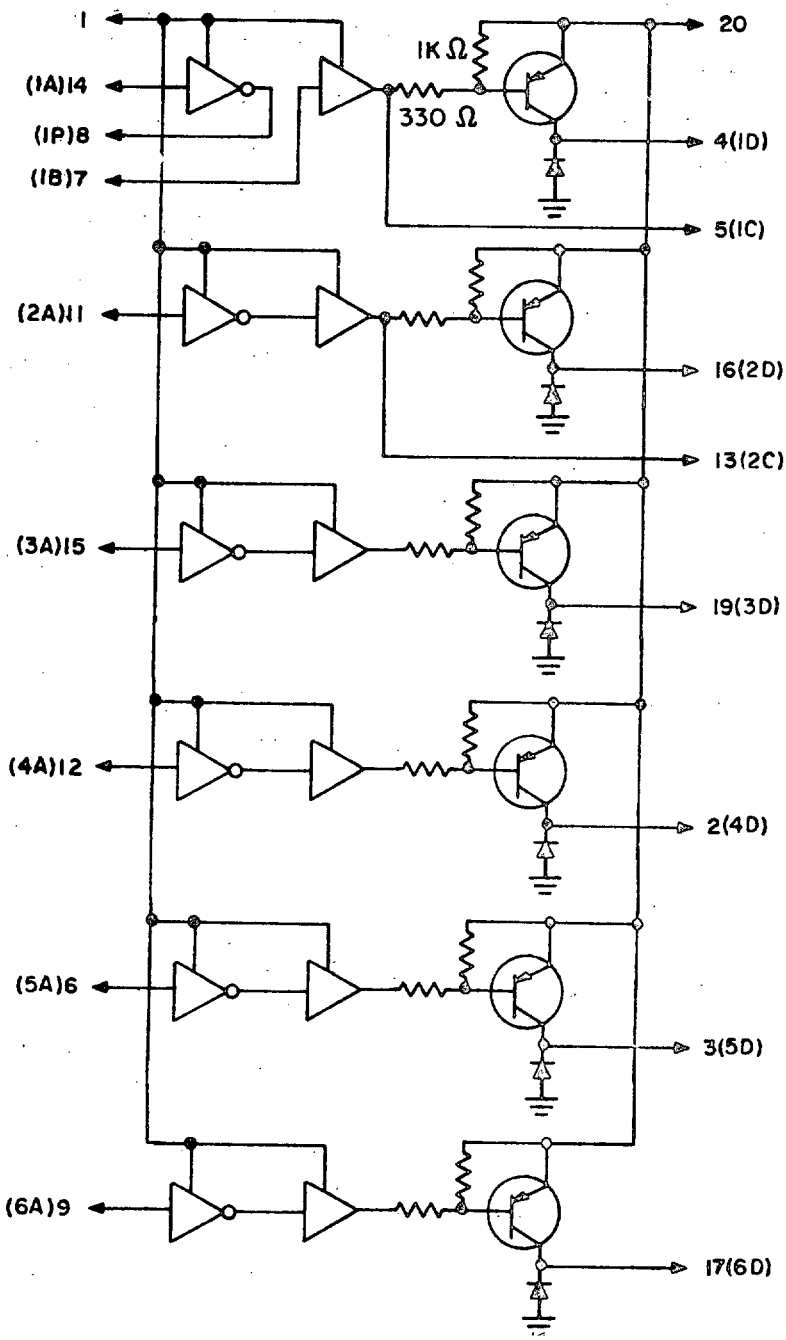
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

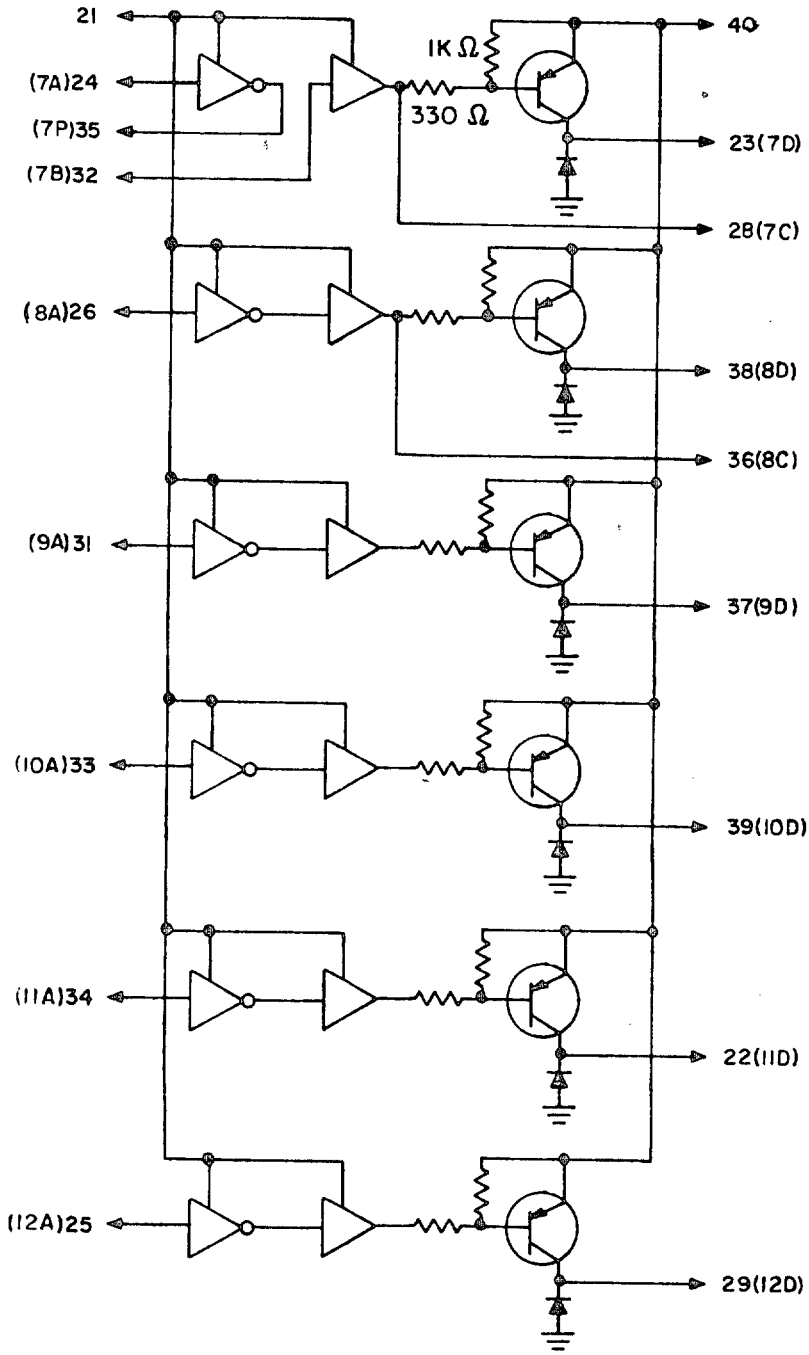
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1, 21	V_{CC} = 5 V ±10% (LOGIC)
10, 30	GND = 0 V REF (LOGIC)
20, 40	V_{CC} = 5 V ±10% (LOAD)
18, 27	GND = 0 V REF (LOAD)

* FIGURE 1. FQB module function diagram, lamp/relay driver.

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* FIGURE 1. FQB module function diagram, lamp/relay driver - Continued.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Frame isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B	Circuit isolation	IV	S	10	∞	1	∞	10	∞	1	∞	MΩ
C1, C2	Integrity	V	S	250	∞	250	∞	250	∞	250	∞	kΩ
C3, C4	Integrity	V	S	1	∞	1	∞	1	∞	1	∞	MΩ
D1-D4	V_{OUT} (low loaded)	VI	S	10	1,000	10	1,000	10	1,000	10	1,000	mV
D5,D6	V_{OUT} (low loaded)	VI	S	10	800	10	800	10	800	10	800	mV
E1-E12	Input breakdown current	VII	S	0	100	0	100	0	100	0	100	µA
E13,E14	Input breakdown current	VII	S	0	1	0	1	0	1	0	1	mA
F1-F12	Lamp transient amplitude	VIII	S	0.0	2.8	0.0	2.8	0.0	2.8	0.0	2.8	V
G1-G12	Output current (low level) maximum stress	IX	X	0	-2	0	-2	0	-2	0	-2	mA
H1,H2	Output short circuit current	X	X	-3	-40	-3	-40	-3	-40	-3	-40	mA
J1-J12	Output clamp diode voltage	XI	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
K1,K2	Input clamp diode voltage	XI	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
L1-L6	Output voltage, low level	XII	X	0	400	0	400	0	400	0	400	mV
M1-M4	Output voltage, high level	XIII	X	4.0	4.5	4.0	4.5	4.0	4.5	4.0	4.5	V
M5,M6	Output voltage, high level	XIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
M7-M20	Output voltage, high level	XIII	X	4.2	4.5	4.2	4.5	4.2	4.5	4.2	4.5	V
N1-N14	Output current, low level	XIV	X	0	-4	0	-4	0	-4	0	-4	µA
P1-P12	Input current, high level	XV	X	0	20	0	20	0	20	0	20	µA
P13,P14	Input current, high level	XV	X	0	40	0	40	0	40	0	40	µA
R1-R12	Input current, low level	XVI	X	-50	-400	-50	-400	-50	-400	-50	-400	µA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R13,R14	Input current, low level	XVI	X	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	mA
S1,S2	Power supply current, V_{CC} logic, high level, D outputs	XVII	S	10	37	10	37	10	37	10	37	mA
S3,S4	Power supply current, V_{CC} logic, low level, D outputs	XVII	S	15	43	15	43	15	43	15	43	mA
S5,S6	Power supply current, V_{CC} load, high level, D outputs	XVII	S	45	100	45	100	45	100	45	100	mA
S7,S8	Power supply current, V_{CC} load, low level, D outputs	XVII	S	0	50	0	50	0	50	0	50	µA
T1-T4	Transition time (t_{TLH}), C_L = 50 pF	XVIII	S	1	35	1	35	1	35	1	35	ns
T5,T6	Transition time (t_{TLH}), C_L = 50 pF	XVIII	S	1	60	1	60	1	60	1	60	ns
T7-T10	Transition time (t_{TLH}), C_L = 150 pF	XVIII	Q	1	50	1	50	1	50	1	50	ns
T11,T12	Transition time (t_{TLH}), C_L = 150 pF	XVIII	Q	1	100	1	100	1	100	1	100	ns
U1-U4	Transition time (t_{THL}), C_L = 50 pF	XVIII	S	1	15	1	15	1	15	1	15	ns
U5,U6	Transition time (t_{THL}), C_L = 50 pF	XVIII	S	1	30	1	30	1	30	1	30	ns
U7-U10	Transition time (t_{THL}), C_L = 150 pF	XVIII	Q	1	25	1	25	1	25	1	25	ns
U11,U12	Transition time (t_{THL}), C_L = 150 pF	XVIII	Q	1	60	1	60	1	60	1	60	ns
V1,V2	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	26	3	26	3	30	3	30	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V3,V4	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	6	116	6	116	6	130	6	130	ns
V5,V6	Propagation delay time (t_{PLH}), C_L = 50 pF	XVIII	S	3	90	3	90	3	99	3	99	ns
V7,V8	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	3	40	3	40	3	40	3	40	ns
V9,V10	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	6	130	6	130	6	145	6	145	ns
V11,V12	Propagation delay time (t_{PLH}), C_L = 150 pF	XVIII	Q	3	100	3	100	3	110	3	110	ns
W1,W2	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	3	35	3	35	3	35	3	35	ns
W3,W4	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	6	120	6	120	6	135	6	135	ns
W5,W6	Propagation delay time (t_{PHL}), C_L = 50 pF	XVIII	S	3	90	3	90	3	99	3	99	ns
W7,W8	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	3	45	3	45	3	45	3	45	ns
W9,W10	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	6	150	6	150	6	170	6	170	ns
W11,W12	Propagation delay time (t_{PHL}), C_L = 150 pF	XVIII	Q	3	115	3	115	3	130	3	130	ns
W13-W24	Propagation delay time (t_{PLH})	XVIII	Q	1	1,000	1	1,000	1	1,000	1	1,000	ns
W25-W36	Propagation delay time (t_{PHL})	XVIII	Q	0.01	10.00	0.01	10.00	0.01	10.00	0.01	10.00	µs
X1-X6	Output crosstalk, low level	XIX	Q	0	400	0	400	0	400	0	400	mV
Y1-Y6	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
Z1,Z2	Power supply transient amplitude	XX	Q	0	500	0	500	0	500	0	500	mV

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
AA1,AA2	Power supply filter capacitance	XXI	S	0.20	0.34	0.20	0.34	0.20	0.34	0.20	0.34	µF
BB1-BB14	Input capacitance	XXII	Q	2	15	2	15	2	15	2	15	pF
CC1-CC12	Saturation voltage, full load	XXIII	Q	10	350	10	350	10	350	10	350	mV
DD1-DD4	Low level input current (I_{IL})	XXIV	Q	8	17	8	17	8	17	8	17	mA
EE1-EE4	High level output current (I_{OH})	XXV	Q	0	15	0	15	0	15	0	15	µA

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, C, D, E, F, G, H, or J are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F, G, H, or J where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through AA and CC through EE shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test conditions AA1 and AA2, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests F through AA and CC through EE shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests F through CC shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/206-1 supersedes part number M28787/206. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits of table I.

Supply voltage

V_{CC} load	5 V dc ±10 percent
V_{CC} logic	5 V dc ±10 percent

Maximum power dissipation
V_{CC} logic power supply current,
V_{CC} logic = 5.5 V dc
(outputs not loaded)

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Circuits 1-6 or 7-12	
Lamp/relay off	43 mA maximum (239 mW maximum)
Lamp/relay on	37 mA maximum (202 mW maximum)
Module	
Lamp/relay off	86 mA maximum (477 mW maximum)
Lamp/relay on	74 mA maximum (403 mW maximum)
V_{CC} load power supply current, V_{CC} logic = 5.5 V dc, V_{CC} load = 5.5 V dc (outputs not loaded)	
Circuits 1-6 or 7-12	
Lamp/relay off	50 µA maximum (0.275 mW maximum)
Lamp/relay on	100 mA maximum (550 mW maximum)
Module	
Lamp/relay off	100 µA maximum (0.55 mW maximum)
Lamp/relay on	200 mA maximum (1.10 W maximum)
Maximum module power dissipation, V_{CC} logic = 5.5 V dc, V_{CC} load = 5.5 V dc, maximum load on outputs	
Circuits 1-6 or 7-12	
Lamp/relay off	239 mW maximum
Lamp/relay on	1,128 mW maximum
Module	
Lamp/relay off	478 mW maximum
Lamp/relay on	2.26 W maximum
Input voltage (inputs A1-A12, B1, B7)	
Low level	0.7 V maximum
High level	2 V minimum, V_{CC} maximum
Input current (inputs A1-A12)	
Low level (V_{IL} = 0.0 V)	-0.4 mA maximum
(V_{IL} = 0.4 V)	-0.36 mA maximum
High level (V_{IH} = 2.7 V)	20 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
Input current (inputs B1, B7)	
Low level (V_{IL} = 0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Input current (inputs C1, C2, C7, C8)	
Low level (V_{IL} = 0 V)	-17 mA maximum
Output voltage (outputs P1, P7)	
Low level (I_{OL} = 2 mA)	0.4 V maximum
High level (I_{OH} = 100 µA)	2.4 V minimum
Output voltage (outputs C1, C2, C7, C8)	
Low level (I_{OL} = 2 mA)	0.4 V maximum
High level (I_{OH} = 100 µA)	4 V minimum
Output drive current (outputs P1, P7) maximum capability	
Low level (V_{OL} = 0.4 V)	2 mA
High level (V_{OH} = 2.4 V)	-100 µA

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Output drive current (outputs C1, C2, C7, C8) maximum capability	
Low level (V_{OL} = 0.4 V)	2 mA
High level (V_{OH} = 4 V)	-100 µA
Output drive current (outputs D1-D12) maximum capability	
Low level (V_{CC} = 4.5 V)	4 µA
Low level (V_{CC} = 3 V)	2 mA
High level (V_{OH} = 4.2 V)	-180 mA
Low level (C input) (I_{OH} = 250 µA)	15 µA
Propagation delay time (A to P)	
t_{PLH} (C_L = 50 pF)	3 ns minimum, 99 ns maximum
(C_L = 150 pF)	3 ns minimum, 110 ns maximum
t_{PHL} (C_L = 50 pF)	3 ns minimum, 99 ns maximum
(C_L = 150 pF)	3 ns minimum, 130 ns maximum
Propagation delay time (B to C)	
t_{PLH} (C_L = 50 pF)	3 ns minimum, 30 ns maximum
(C_L = 150 pF)	3 ns minimum, 40 ns maximum
t_{PHL} (C_L = 50 pF)	3 ns minimum, 35 ns maximum
(C_L = 150 pF)	3 ns minimum, 45 ns maximum
Propagation delay time (A to C)	
t_{PLH} (C_L = 50 pF)	6 ns minimum, 130 ns maximum
(C_L = 150 pF)	6 ns minimum, 145 ns maximum
t_{PHL} (C_L = 50 pF)	6 ns minimum, 135 ns maximum
(C_L = 150 pF)	6 ns minimum, 170 ns maximum

* 6.5 Module function. This module contains 12 independent circuits with the following functional capability.

a. All 12 circuits have the capability of driving a single 5 V, 180 mA (steady-state) lamp or up to two 5 V, 90 mA relay coils with a high-level input. Peak in-rush current is limited to 1.1 A maximum by the driver circuitry.

b. Circuits 1 and 7 have the capability of driving a single 5 V, 180 mA lamp or up to two 5 V, 90 mA relay coils with a low-level input.

c. Circuits 1, 2, 7, and 8 have the capability of being used as inverters having low-power TTL inputs and either; (1) open collector TTL outputs requiring external pullup resistors (V_{CC} (load) not connected), or; (2) buffer outputs with pullup resistor included in module circuitry (V_{CC} (load) is connected to V_{CC} (logic)).

d. Circuits 1 and 7 have the capability of being used as inverters having low-power TTL inputs and low-power TTL outputs.

e. Circuits 2 and 8 have the capability of being used as buffers having regular TTL inputs with outputs as specified in c(1) and c(2) above.

6.5.1 Input to output relationship. All lamp/relay driver input to output relationships are represented in table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Lamp and logic power supply separation. If more than one lamp is driven by a single supply, the V_{CC} (logic) and V_{CC} (load) module pins should have separate supplies. V_{CC} (logic) voltage shall be applied to the module before V_{CC} (load) voltage is applied to the module.

6.5.4 Separating logic functions and lamp/relay drivers. If lamps or relays are being driven by a driver circuits 1-6 or 7-12, then the logic functions of circuits 1 and 2 or 7 and 8, respectively, may not be used.

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6.5.5 Driver output leakage at low V_{CC} (logic). If lamps or relays are being commanded "off" (logic "0" output) and V_{CC} (logic) dips to 3 V dc, the output leakage current shall not exceed 2 mA through each "off" lamp or relay load.

6.6 Changes for previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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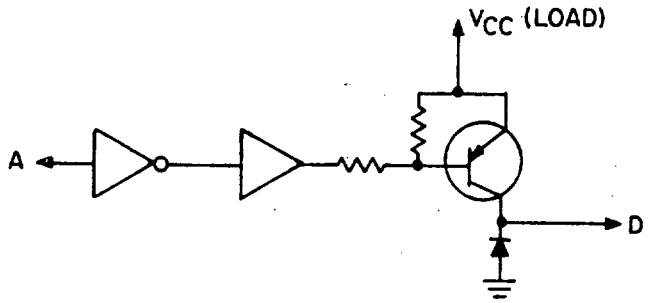
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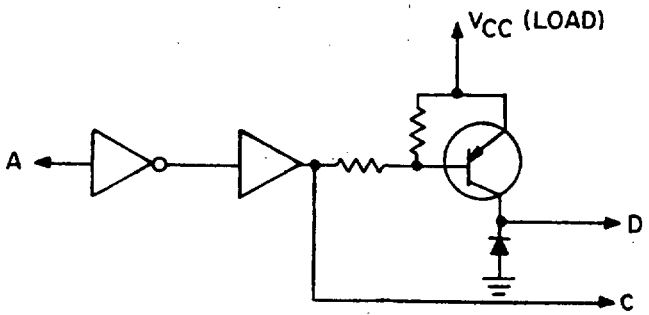
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*TABLE II. Input to output relationship. 1/,2/

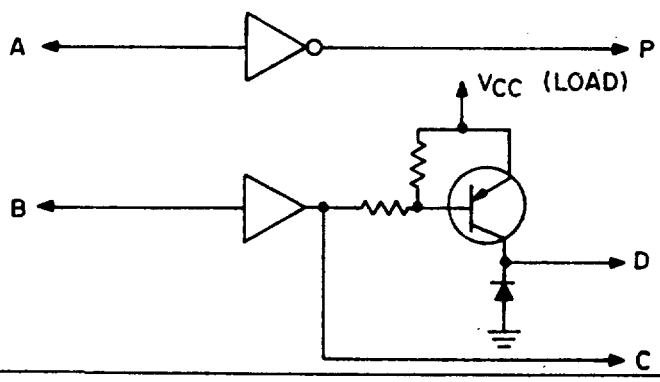
CIRCUITS 1-12	INPUTS	OUTPUTS
	A	D
	L	L (LAMP OFF)
	H	H (LAMP ON)



CIRCUITS 1, 2, 7, AND 8	INPUTS	OUTPUTS	
	A	C	D
	L	H	L (LAMP OFF)
	H	L	H (LAMP ON)

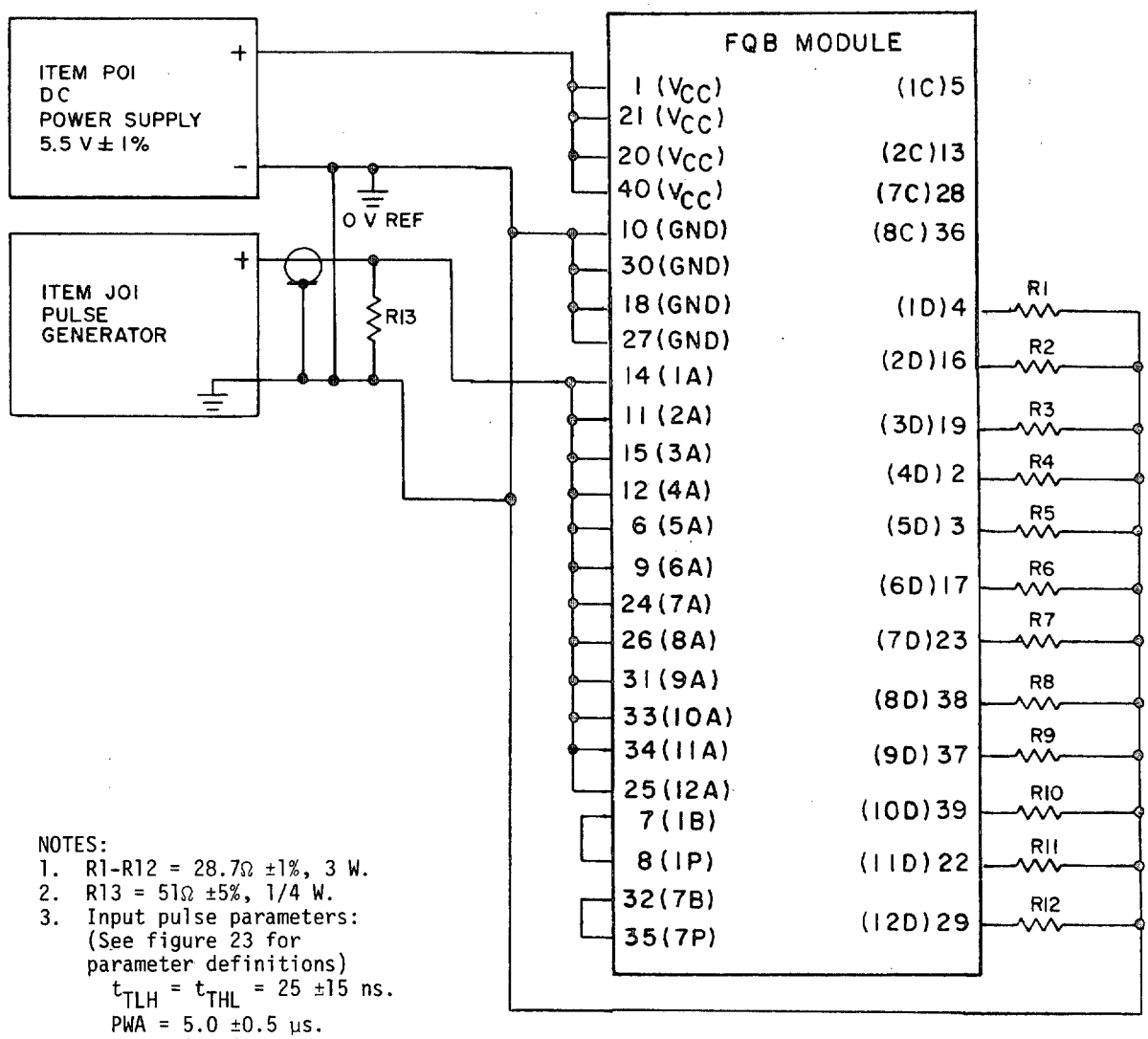


CIRCUITS 1 AND 7	INPUTS		OUTPUTS		
	A	B	C	D	P
	H	L	L	H (LAMP ON)	L
	L	H	H	L (LAMP OFF)	H



1/ H = High level.
2/ L = Low level.

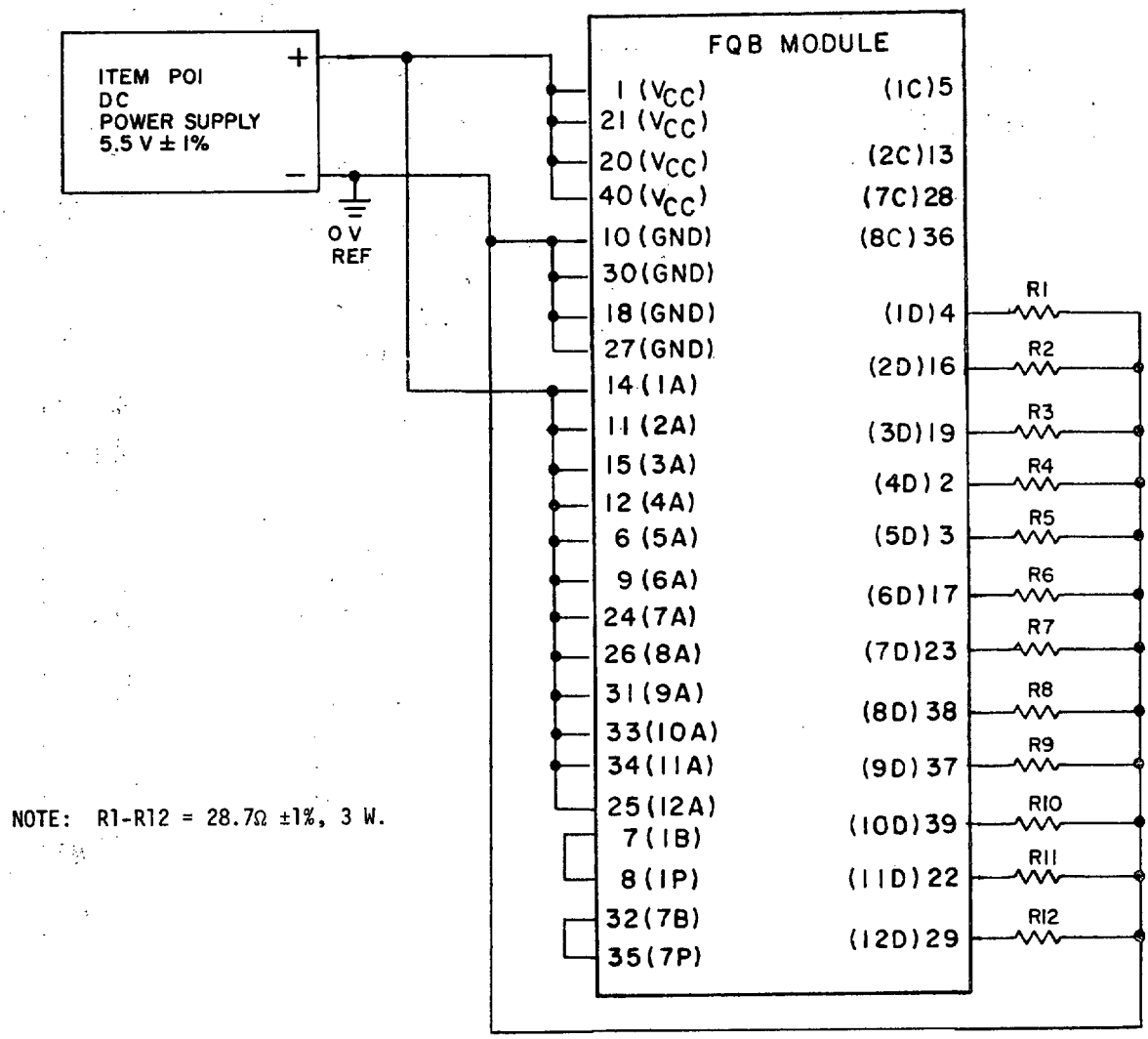
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NOTES:
1. R1-R12 = 28.7Ω ±1%, 3 W.
2. R13 = 51Ω ±5%, 1/4 W.
3. Input pulse parameters:
   (See figure 23 for
   parameter definitions)
   $t_{TLH} = t_{THL}$ = 25 ±15 ns.
   PWA = 5.0 ±0.5 µs.
   PRP = 20 ±2 µs.
   PA = 3.0 ±0.1 V.

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R12 = 28.7Ω ±1%, 3 W.

FIGURE 3. Component temperature test circuit.

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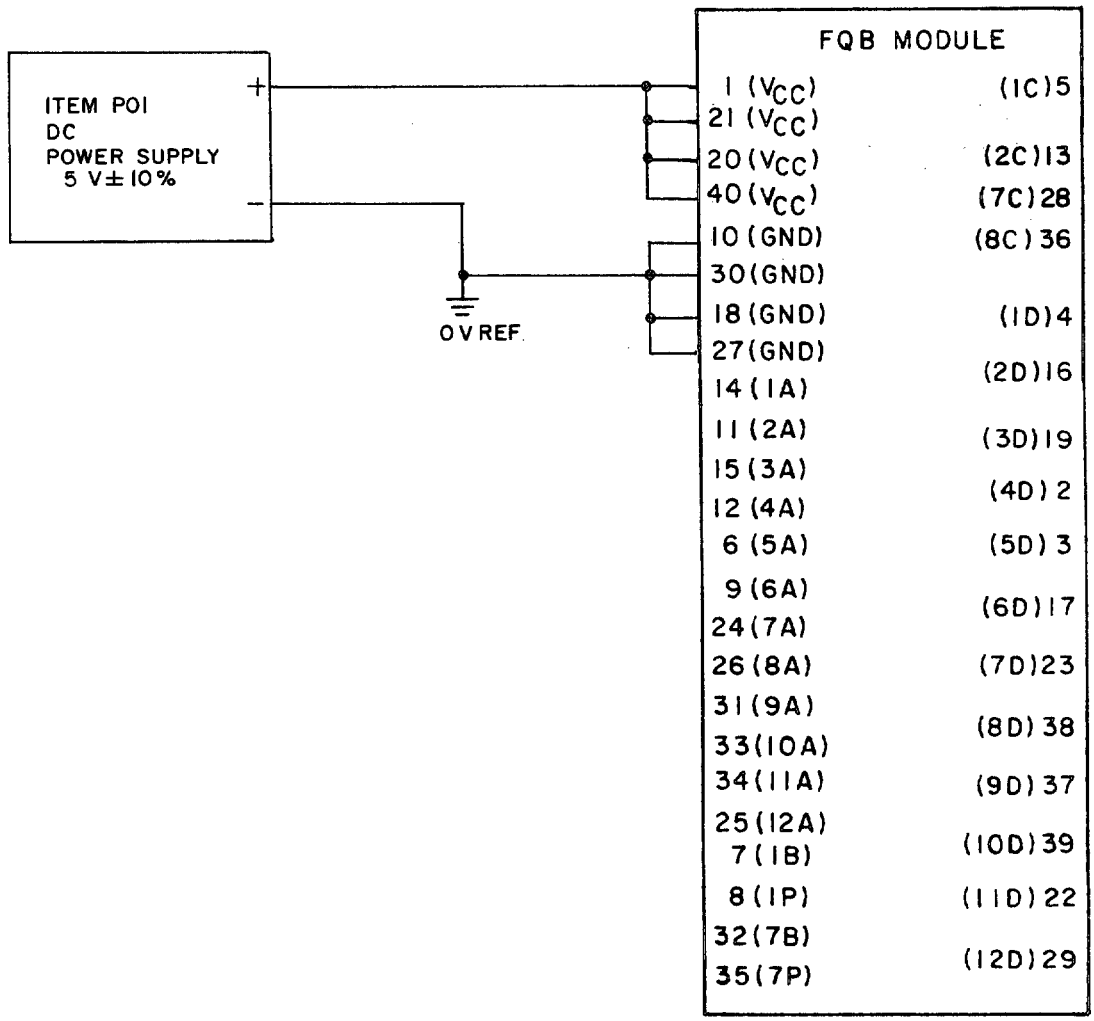


FIGURE 4. Vibration test circuit.

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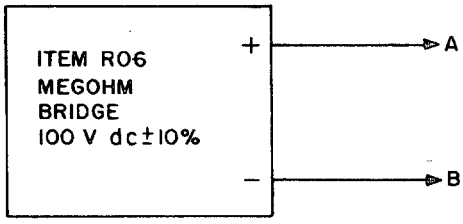
TABLE III. Frame isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Frame isolation	C	Module frame	10	∞	MΩ	5

TABLE IV. Circuit isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B	Circuit isolation	1	21	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

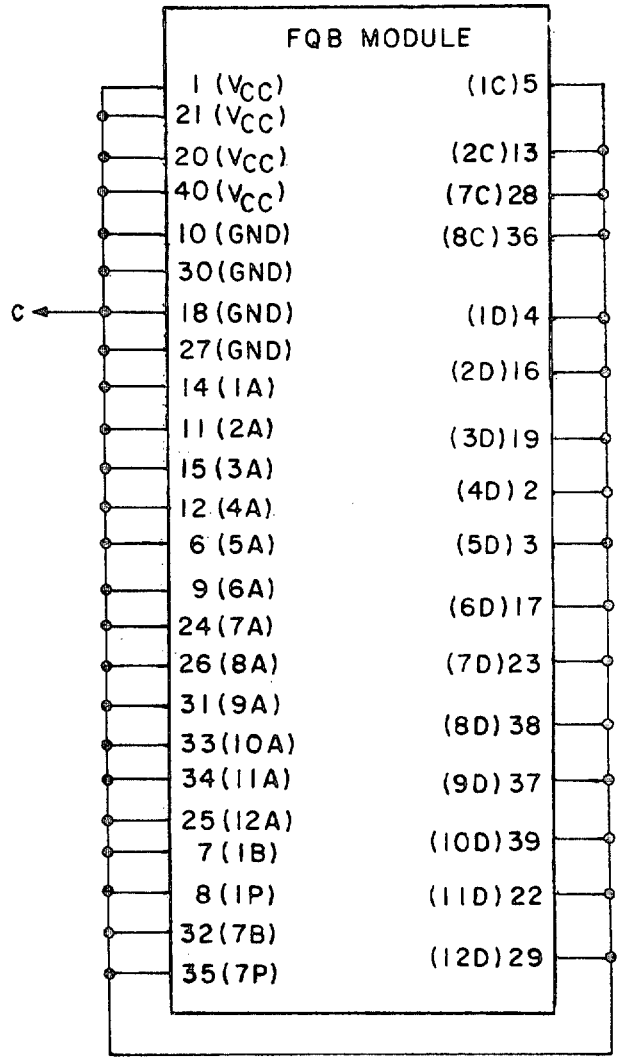
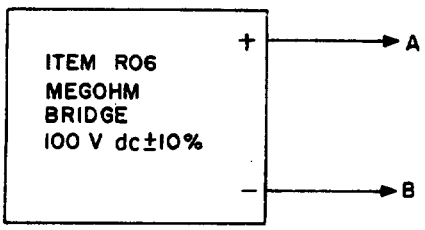


FIGURE 5. Frame isolation test circuit for table III.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

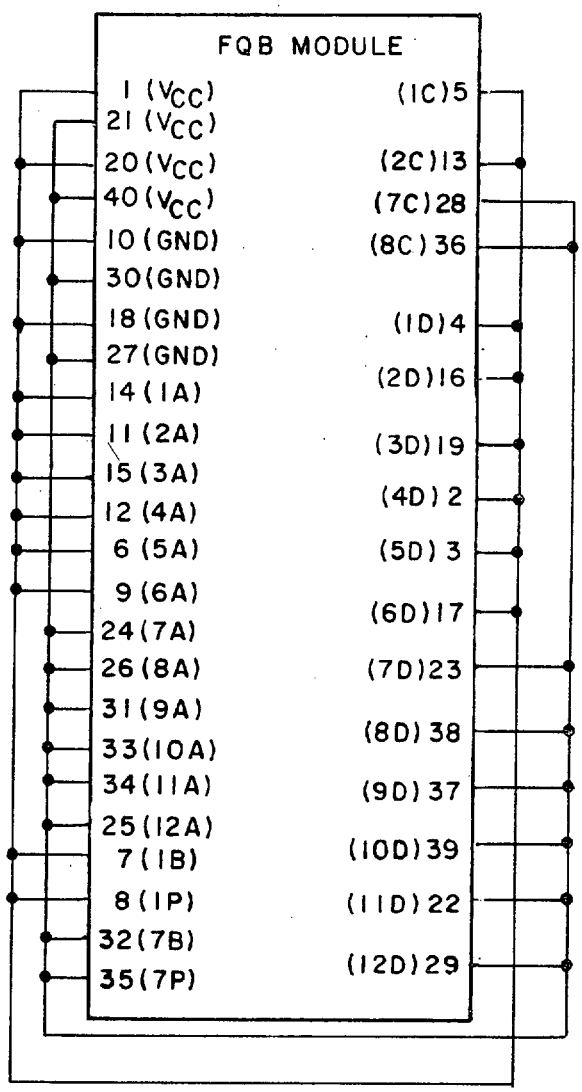


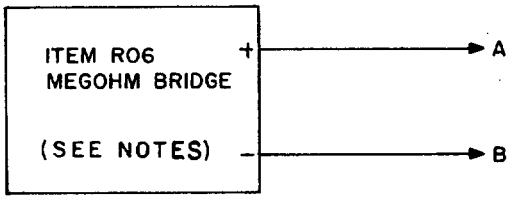
FIGURE 6. Circuit isolation test circuit for table IV.

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*TABLE V. Integrity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Integrity	20	1	250	∞	kΩ	7
C2	"	40	21	250	"	kΩ	"
C3	"	10	18	1	"	MΩ	"
C4	"	30	27	1	"	MΩ	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. 10 V dc for test conditions C1 and C2.
2. 50 V dc for test conditions C3 and C4.

FQB MODULE	
1 (V_{CC})	(1C)5
21 (V_{CC})	
20 (V_{CC})	(2C)13
40 (V_{CC})	(7C)28
10 (GND)	(8C)36
30 (GND)	
18 (GND)	(1D)4
27 (GND)	
14 (1A)	(2D)16
11 (2A)	(3D)19
15 (3A)	
12 (4A)	(4D)2
6 (5A)	(5D)3
9 (6A)	
24 (7A)	(6D)17
26 (8A)	(7D)23
31 (9A)	
33 (10A)	(8D)38
34 (11A)	(9D)37
25 (12A)	
7 (1B)	(10D)39
8 (1P)	(11D)22
32 (7B)	
35 (7P)	(12D)29

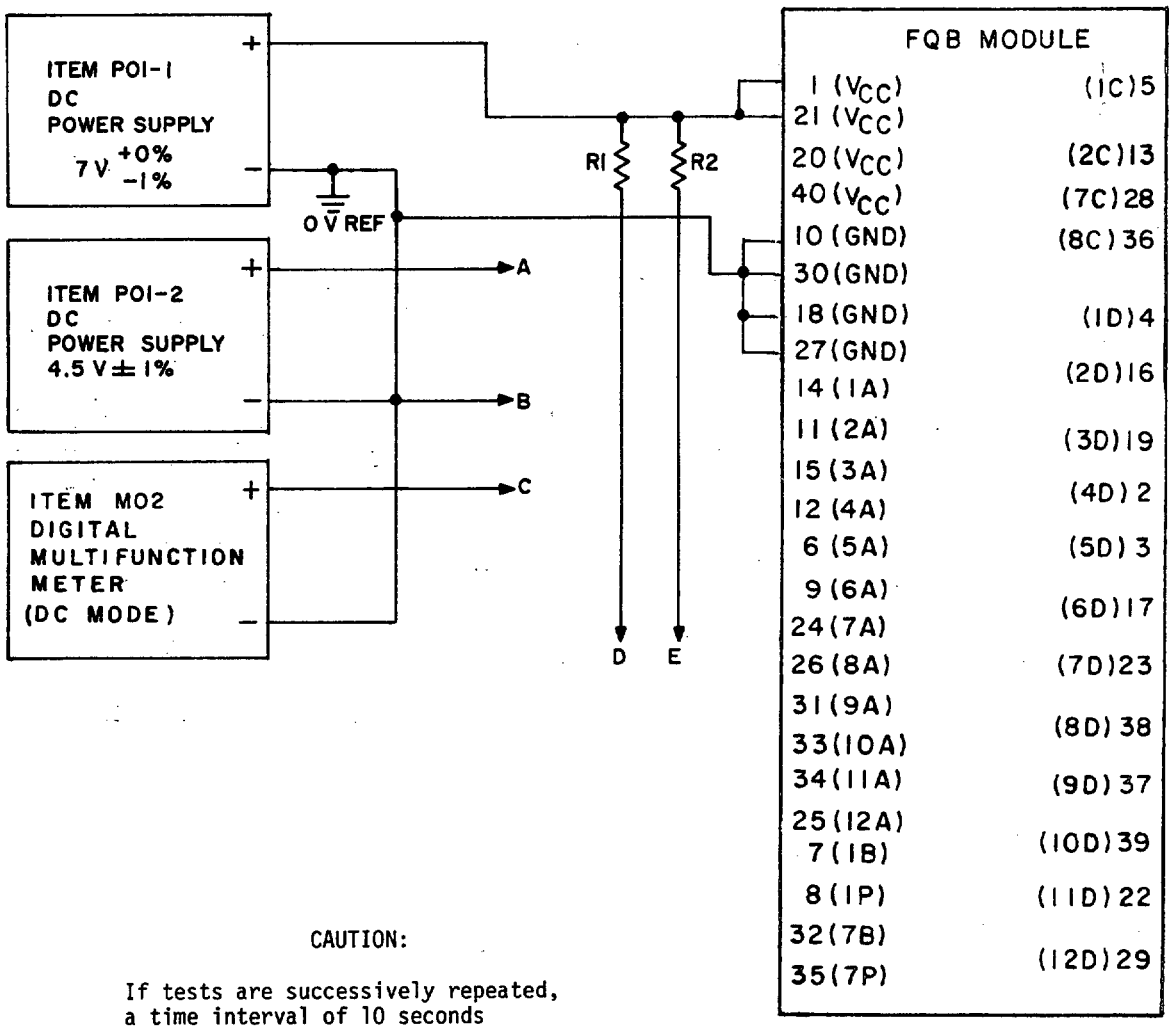
*FIGURE 7. Integrity test circuit for table V.

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*TABLE VI. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
D1	1C V_{OUT} (low loaded)	Open	7	5	Open	5	10	1,000	mV	8
D2	2C "	11	Open	13	"	13	"	"	"	"
D3	7C "	Open	32	28	"	28	"	"	"	"
D4	8C "	26	Open	36	"	36	"	"	"	"
D5	1P "	14	"	8	8	Open	"	800	"	"
D6	7P "	24	"	35	35	Open	"	800	"	"

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CAUTION:

If tests are successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

NOTES:
1. R1 = 2.55 kΩ ±1%, 1/4 W.
2. R2 = 160Ω ±1%, 1/2 W.

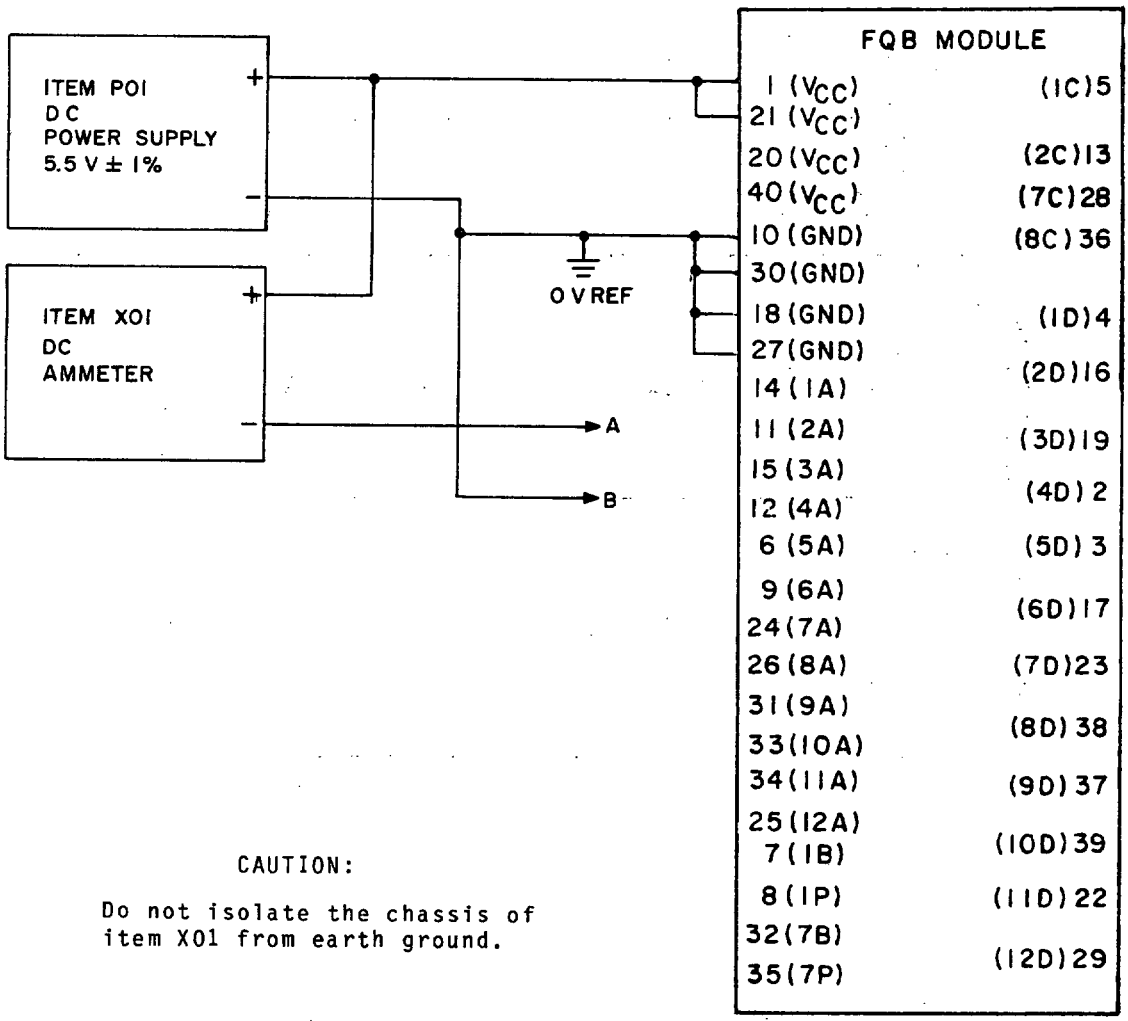
*FIGURE 8. V_{OUT} (low-loaded) test circuit for table VI.

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TABLE VII. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1A Input breakdown current	14	7,11,15,12,6,9,24,32,26, 31,33,34,25	0	100	µA	9
E2	2A "	11	14,7,15,12,6,9,24,32,26, 31,33,34,25	"	"	"	"
E3	3A "	15	14,7,11,12,6,9,24,32,26, 31,33,34,25	"	"	"	"
E4	4A "	12	14,7,11,15,6,9,24,32,26, 31,33,34,25	"	"	"	"
E5	5A "	6	14,7,11,15,12,9,24,32,26, 31,33,34,25	"	"	"	"
E6	6A "	9	14,7,11,15,12,6,24,32,26, 31,33,34,25	"	"	"	"
E7	7A "	24	14,7,11,15,12,6,9,32,26, 31,33,34,25	"	"	"	"
E8	8A "	26	14,7,11,15,12,6,9,24,32, 31,33,34,25	"	"	"	"
E9	9A "	31	14,7,11,15,12,6,9,24,32, 26,33,34,25	"	"	"	"
E10	10A "	33	14,7,11,15,12,6,9,24,32, 26,31,34,25	"	"	"	"
E11	11A "	34	14,7,11,15,12,6,9,24,32, 26,31,33,25	"	"	"	"
E12	12A "	25	14,7,11,15,12,6,9,24,32, 26,31,33,34	"	"	"	"
E13	1B "	7	14,11,15,12,6,9,24,32, 26,31,33,34,25	"	1	mA	"
E14	7B "	32	14,7,11,15,12,6,9,24,26, 31,33,34,25	"	1	mA	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

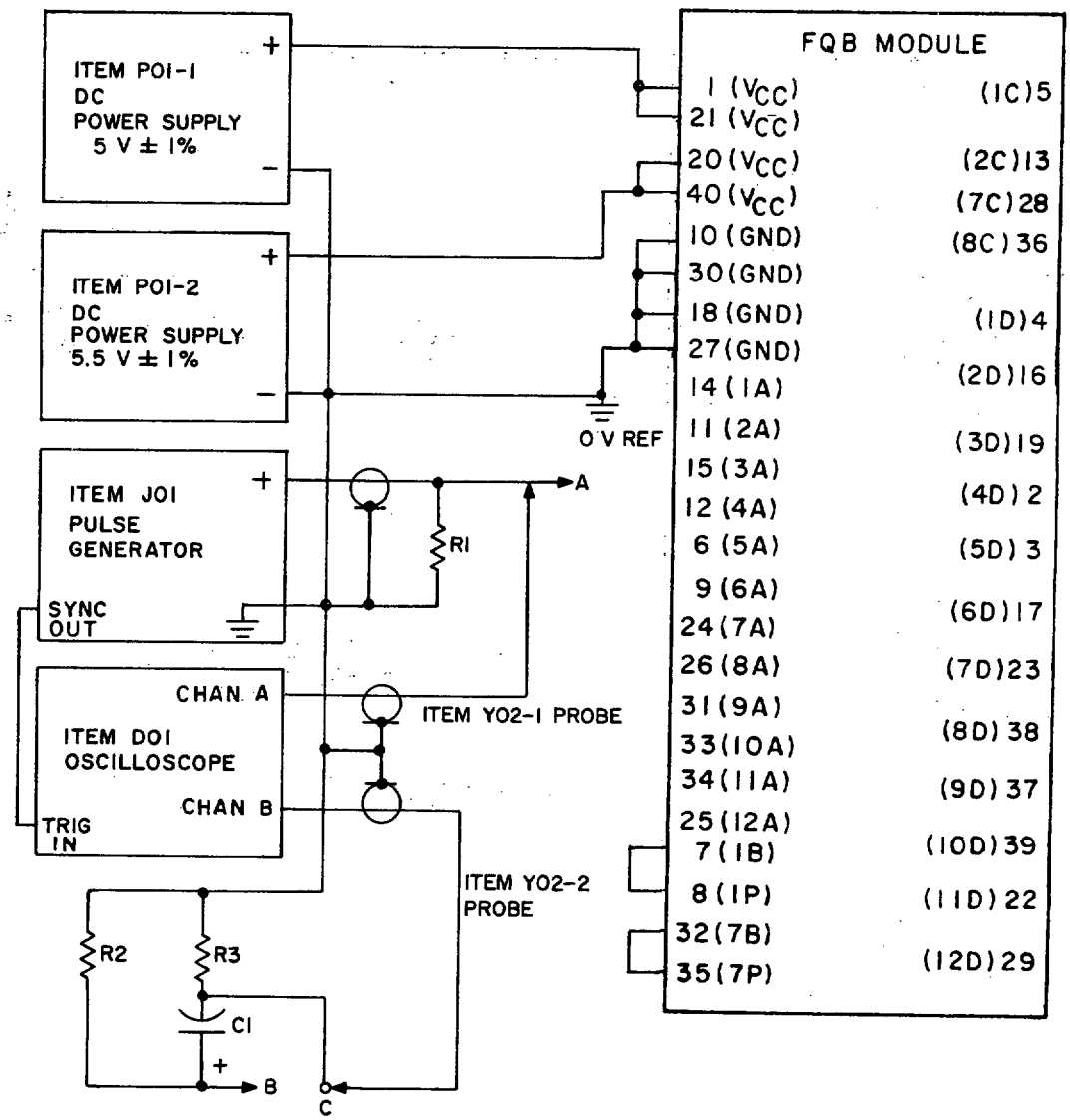
FIGURE 9. Input breakdown current test circuit for table VII.

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*TABLE VIII. Lamp transient amplitude test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	1D	Lamp transient amplitude	14	4	0.0	2.8	V	10,11
F2	2D	"	11	16	"	"	"	"
F3	3D	"	15	19	"	"	"	"
F4	4D	"	12	2	"	"	"	"
F5	5D	"	6	3	"	"	"	"
F6	6D	"	9	17	"	"	"	"
F7	7D	"	24	23	"	"	"	"
F8	8D	"	26	38	"	"	"	"
F9	9D	"	31	37	"	"	"	"
F10	10D	"	33	39	"	"	"	"
F11	11D	"	34	22	"	"	"	"
F12	12D	"	25	29	"	"	"	"

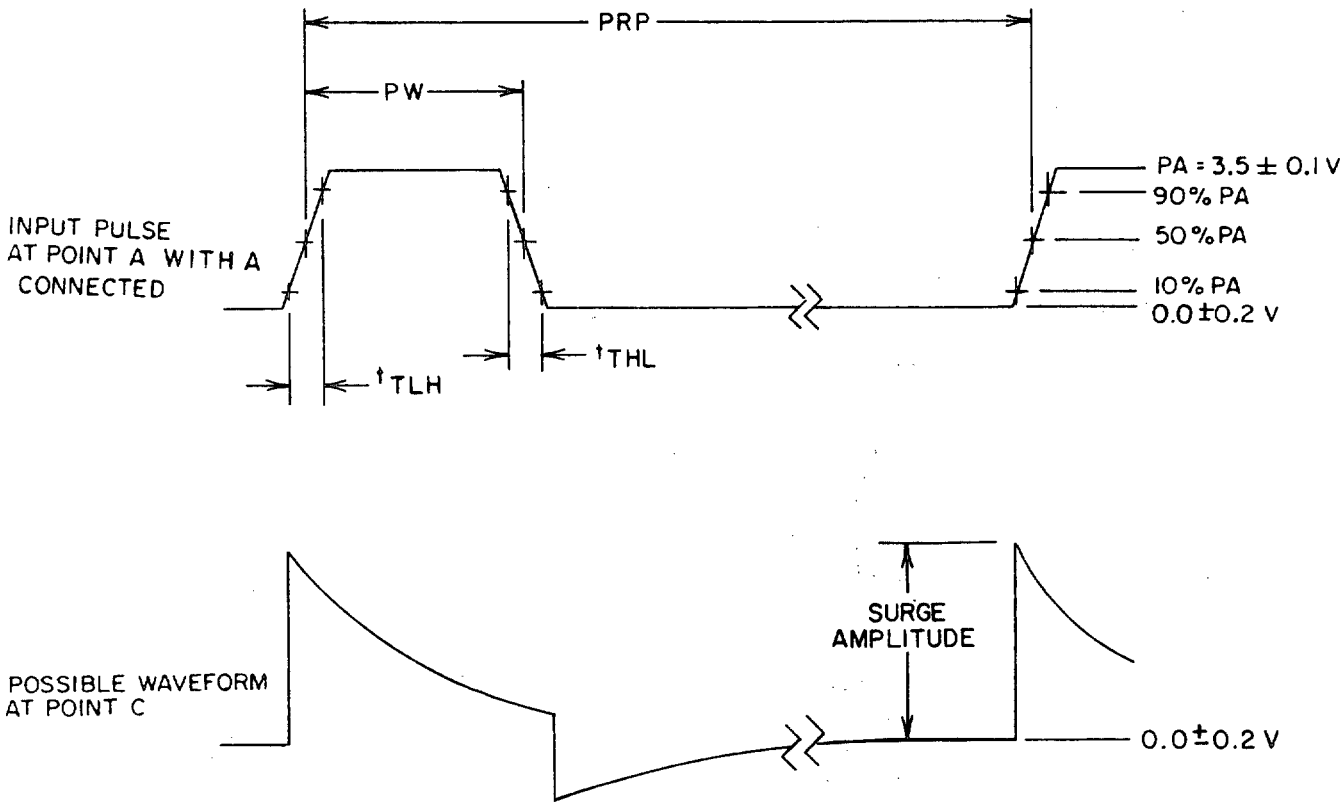
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:39Z
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NOTES:
1. R1 = 50Ω ±5%, 1/4 W.
2. R2 = 28.7Ω ±1%, 3 W.
3. R3 = 2.5Ω ±1%, 5 W (nonconductive).
4. C1 = 1,000 µF ±20%, 6 V dc.
5. Total lead length of B to capacitor C1 plus C1 to R3 to 0 V REF shall be less than 4 inches (101.6 mm).
6. This test verifies that the output will not supply more than 1A of surge current.

FIGURE 10. Lamp transient amplitude test circuit for table VIII.

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NOTE: Input pulse parameters:
PRP = 300 ±30 ms.
PW = 15.0 ±1.5 ms.
$t_{TLH} = t_{THL} = 10 \pm 2$ µs.

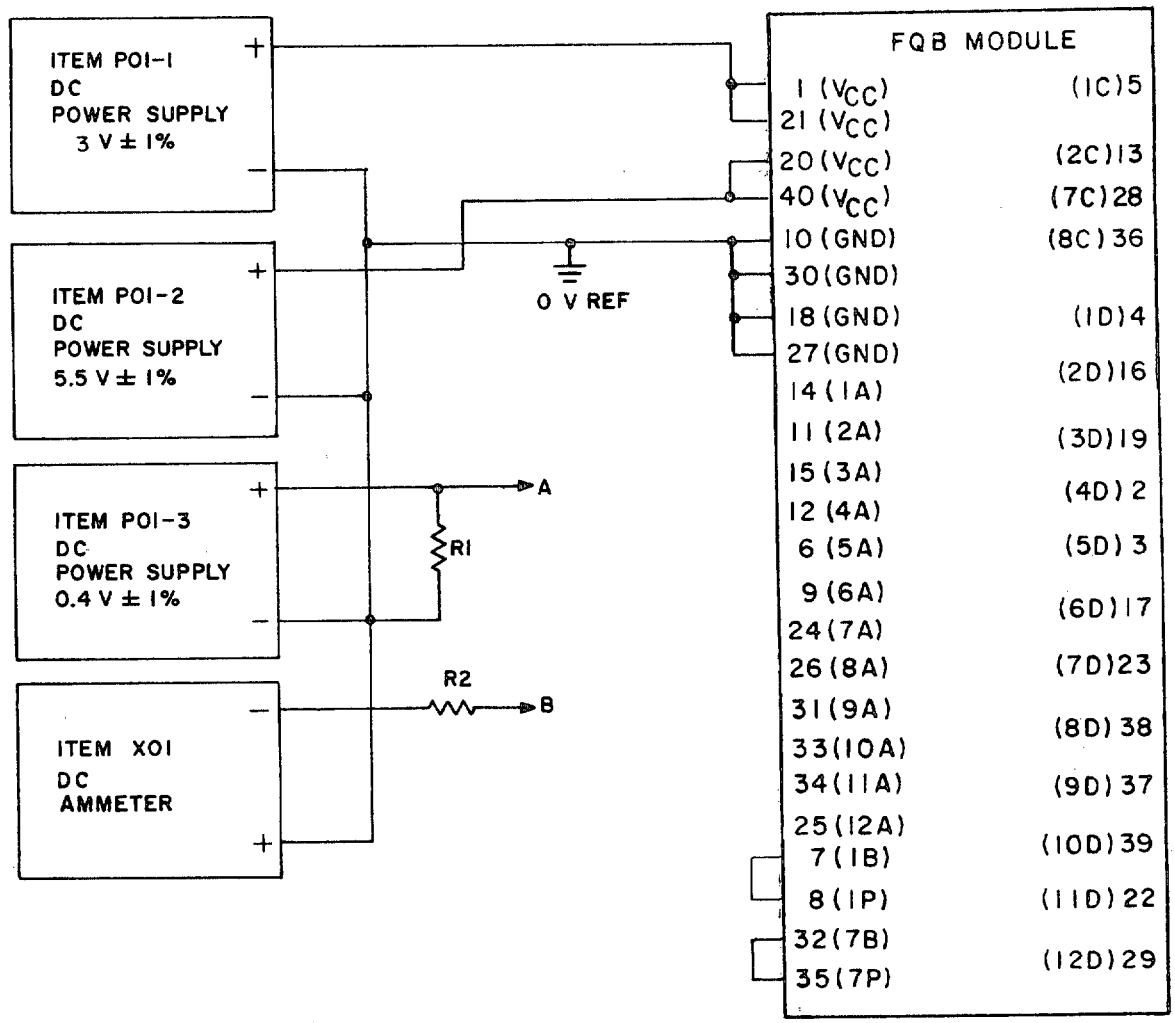
FIGURE 11. Lamp transient amplitude test circuit waveforms for table VIII.

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*TABLE IX. Output current (low-level) maximum stress test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G1	1D	Output current (low level) maximum stress	14	4	0	-2	mA	12
G2	2D	"	11	16	"	"	"	"
G3	3D	"	15	19	"	"	"	"
G4	4D	"	12	2	"	"	"	"
G5	5D	"	6	3	"	"	"	"
G6	6D	"	9	17	"	"	"	"
G7	7D	"	24	23	"	"	"	"
G8	8D	"	26	38	"	"	"	"
G9	9D	"	31	37	"	"	"	"
G10	10D	"	33	39	"	"	"	"
G11	11D	"	34	22	"	"	"	"
G12	12D	"	25	29	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. R2 = 28.7 ±1%, 1 W.

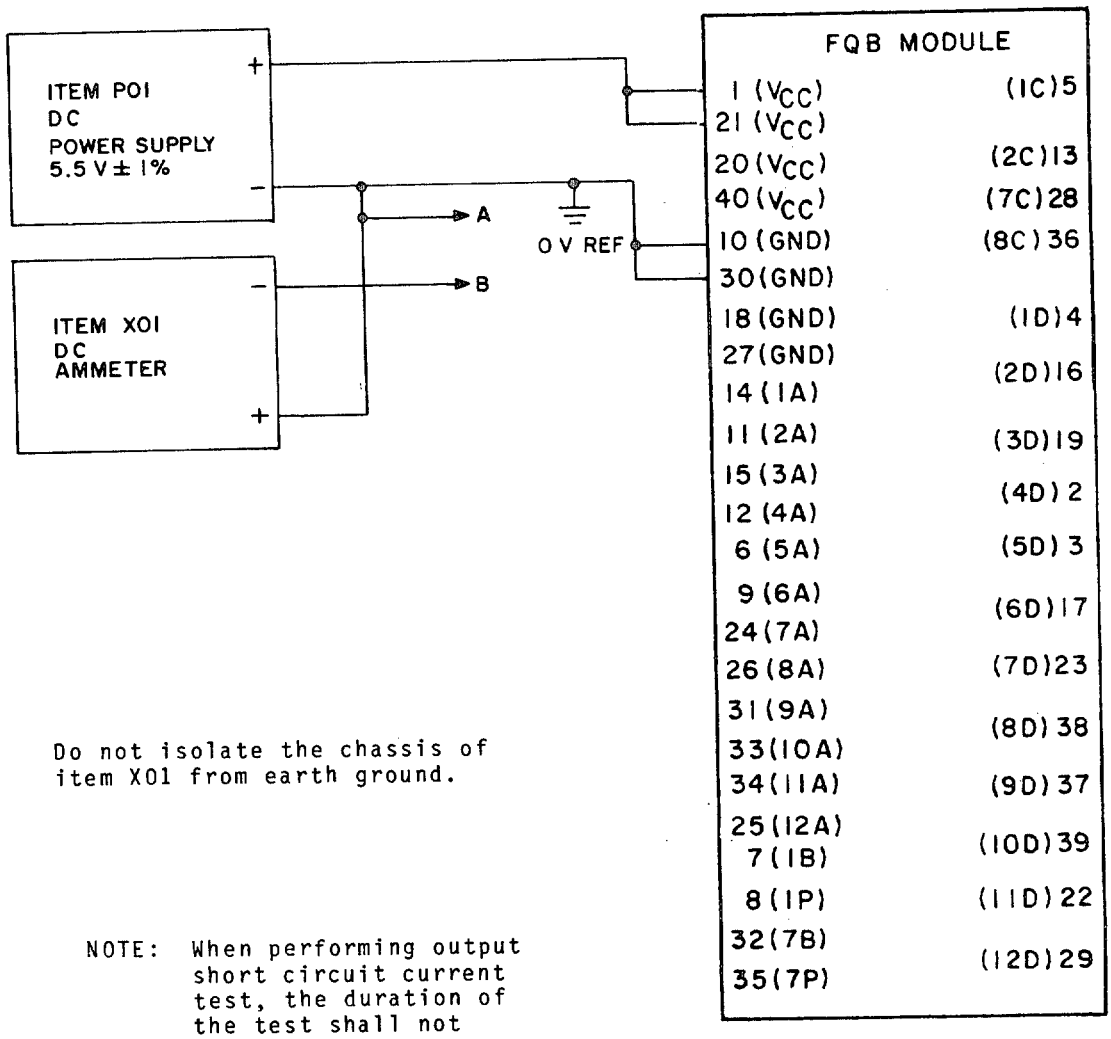
*FIGURE 12. Output current (low-level) maximum stress test circuit for table IX.

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	1P Output short circuit current	14	8	-3	-40	mA	13
H2	7P Output short circuit current	24	35	-3	-40	mA	13

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Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing output short circuit current test, the duration of the test shall not exceed five seconds.

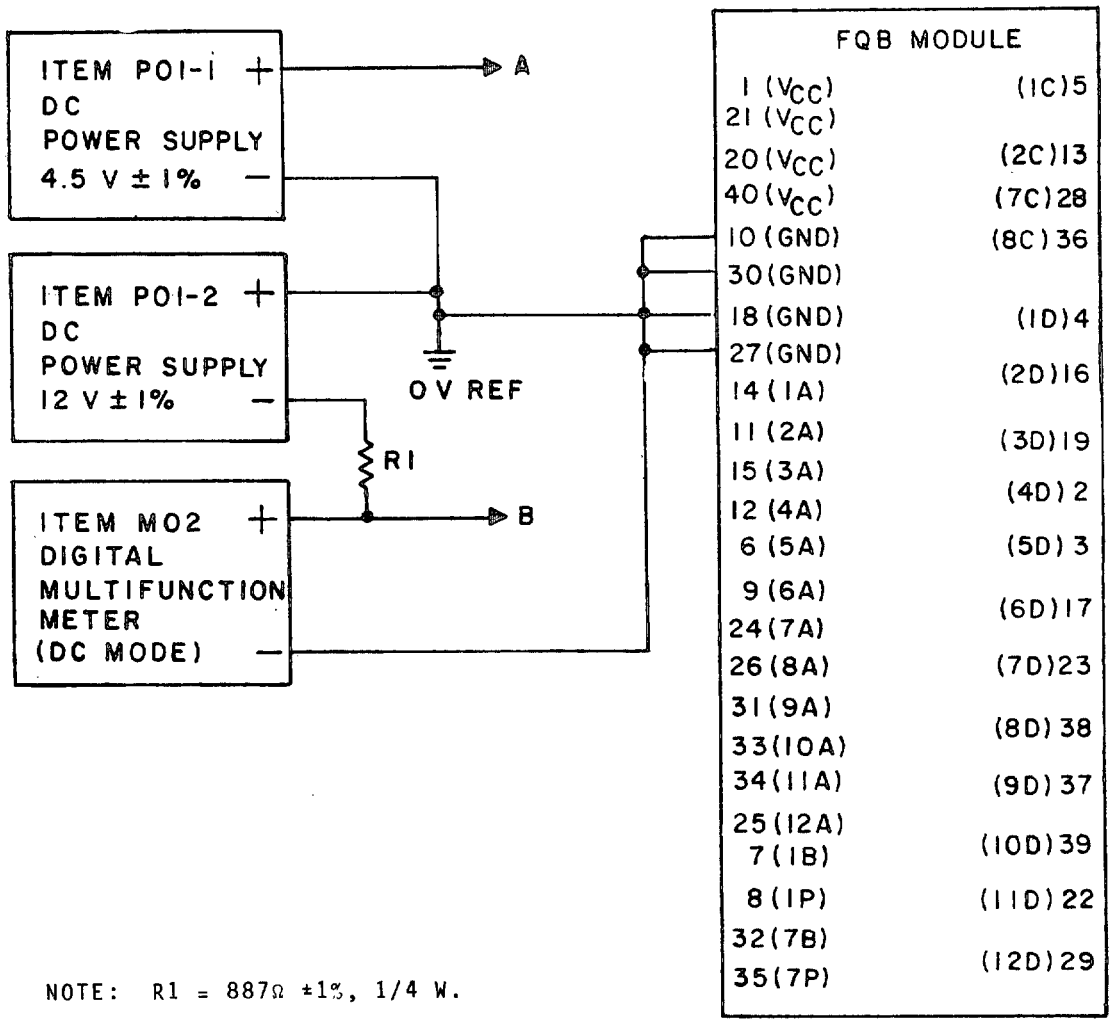
FIGURE 13. Output short circuit current test circuit for table X.

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*TABLE XI. Output and input clamp diode voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1D Output clamp diode voltage	Open	4	-0.4	-1.2	V	14
J2	2D "	"	16	"	"	"	"
J3	3D "	"	19	"	"	"	"
J4	4D "	"	2	"	"	"	"
J5	5D "	"	3	"	"	"	"
J6	6D "	"	17	"	"	"	"
J7	7D "	"	23	"	"	"	"
J8	8D "	"	38	"	"	"	"
J9	9D "	"	37	"	"	"	"
J10	10D "	"	39	"	"	"	"
J11	11D "	"	22	"	"	"	"
J12	12D "	"	29	"	"	"	"
K1	1B Input clamp diode voltage	1	7	"	-1.5	"	"
K2	7B Input clamp diode voltage	21	32	"	-1.5	"	"

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NOTE: R1 = 887Ω ±1%, 1/4 W.

*FIGURE 14. Output and input clamp diode voltage test circuit for table XI.

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*TABLE XII. Output voltage (low-level) test.

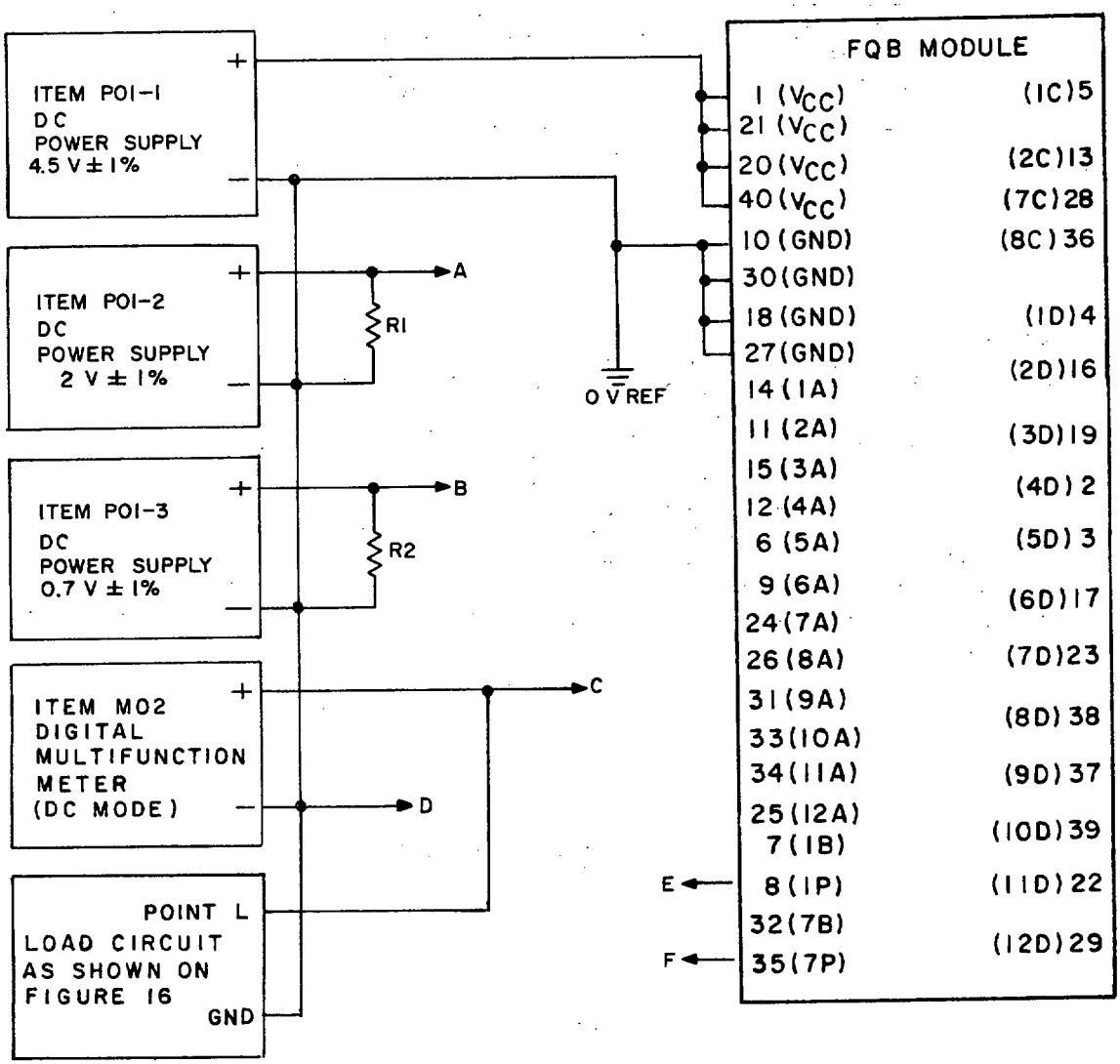
Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L1	1C Output voltage, low level	Open	7	5	Open	Open	Open	0	400	mV	15
L2	2C "	11	Open	13	"	"	"	"	"	"	"
L3	7C "	Open	32	28	"	"	"	"	"	"	"
L4	8C "	26	Open	36	"	"	"	"	"	"	"
L5	1P "	14	"	8	"	"	"	"	"	"	"
L6	7P "	24	"	35	"	"	"	"	"	"	"

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*TABLE XIII. Output voltage (high-level) test.

Test condition	Parameter	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M1	1C Output voltage, high level	7	Open	5	11,24,26,32,14	Open	Open	4.0	4.5	V	15
M2	2C "	Open	11	13	7,14,24,26,32	"	"	"	"	"	"
M3	7C "	32	Open	28	7,14,11,26,24	"	"	"	"	"	"
M4	8C "	Open	26	36	7,14,11,24,32	"	"	"	"	"	"
M5	1P "	"	14	8	7,11,24,26,32	"	"	2.4	"	"	"
M6	7P "	"	24	35	7,11,14,26,32	"	"	2.4	"	"	"
M7	1D "	14	Open	4	11,15,12,6,9	7	32	4.2	"	"	"
M8	2D "	11	"	16	14,15,12,6,9	"	"	"	"	"	"
M9	3D "	15	"	19	14,11,12,6,9	"	"	"	"	"	"
M10	4D "	12	"	2	14,11,15,6,9	"	"	"	"	"	"
M11	5D "	6	"	3	14,11,15,12,9	"	"	"	"	"	"
M12	6D "	9	"	17	14,11,15,12,6	"	"	"	"	"	"
M13	7D "	24	"	23	26,31,33,34,25	"	"	"	"	"	"
M14	8D "	26	"	38	24,31,33,34,25	"	"	"	"	"	"
M15	9D "	31	"	37	24,26,33,34,25	"	"	"	"	"	"
M16	10D "	33	"	39	24,26,31,34,25	"	"	"	"	"	"
M17	11D "	34	"	22	24,26,31,33,25	"	"	"	"	"	"
M18	12D "	25	"	29	24,26,31,33,34	"	"	"	"	"	"
M19	1D "	Open	7	4	Open	Open	Open	"	"	"	"
M20	7D "	Open	32	23	Open	Open	Open	"	"	"	"

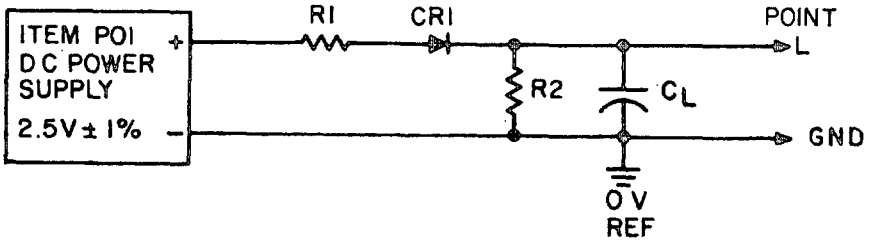
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/4 W.
2. Use load A of figure 16 for test conditions L1-L6 and M1-M6.
3. Use load B of figure 16 for test conditions M7-M20.

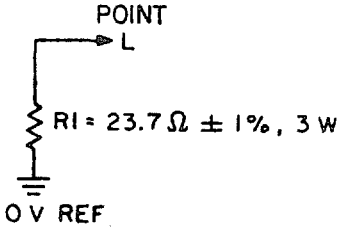
*FIGURE 15. Output voltages (low-level and high-level) test circuit for tables XII and XIII.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 2 mA ±5% out of point L with point L held at 0.4 V ±1%.
3. R2 = 40.2 kΩ ±1%, 1/8 W (for pins 5, 13, 28, and 36).
   R2 = 24.3 kΩ ±1%, 1/8 W (for pins 8 and 35).
4. For test conditions T1-T6, U1-U6, V1-V6, W1-W6, and Y1-Y6: C_L = 50 ±5 PF. (including probe and fixture capacitance).
5. For test conditions T7-T12, U7-U12, V7-V12, and W7-W12: C_L = 150 ±5 pF (including probe and fixture capacitance).
6. Capacitor C_L is not required for dc electrical test conditions L1-L6 and M1-M6.
7. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

LOAD A (LOGIC LOAD)



LOAD B (LOGIC LOAD)

NOTE: Load B is for test conditions M7-M20.

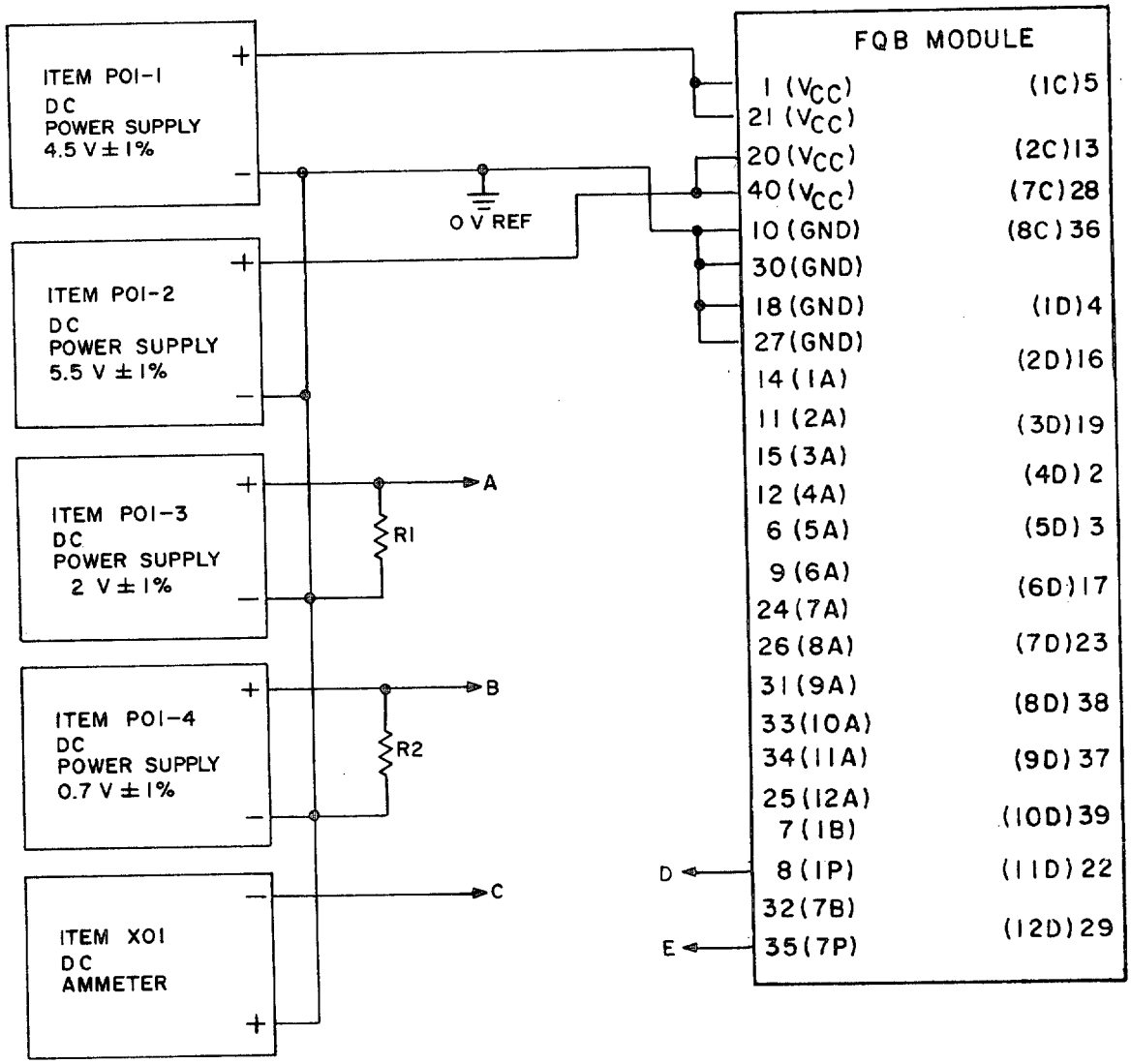
FIGURE 16. FQB logic load circuits.

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*TABLE XIV. Output current (low-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	1D Output current, low level	11,15,12,6,9,24, 26,31,33,34,25	14	4	7	32	0	-4	µA	17
N2	2D "	14,15,12,6,9,24, 26,31,33,34,25	11	16	"	"	"	"	"	"
N3	3D "	14,11,12,6,9,24, 26,31,33,34,25	15	19	"	"	"	"	"	"
N4	4D "	14,11,15,6,9,24, 26,31,33,34,25	12	2	"	"	"	"	"	"
N5	5D "	14,11,15,12,9,24, 26,31,33,34,25	6	3	"	"	"	"	"	"
N6	6D "	14,11,15,12,6,24, 26,31,33,34,25	9	17	"	"	"	"	"	"
N7	7D "	14,11,15,12,6,9, 26,31,33,34,25	24	23	"	"	"	"	"	"
N8	8D "	14,11,15,12,6,9, 24,31,33,34,25	26	38	"	"	"	"	"	"
N9	9D "	14,11,15,12,6,9, 24,26,33,34,25	31	37	"	"	"	"	"	"
N10	10D "	14,11,15,12,6,9, 24,26,31,34,25	33	39	"	"	"	"	"	"
N11	11D "	14,11,15,12,6,9, 24,26,31,33,25	34	22	"	"	"	"	"	"
N12	12D "	14,11,15,12,6,9, 24,26,31,33,34	25	29	"	"	"	"	"	"
N13	1D "	7,14,11,15,12,6,9, 24,26,31,33,34,25	---	4	---	---	"	"	"	"
N14	7D "	32,14,11,15,12,6, 9,24,26,31,33,34, 25	---	23	---	---	"	"	"	"

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CAUTION:
Do not isolate the chassis of item XO1 from earth ground.

NOTE: R1 = R2 = 51Ω ±5%, 1/4 W.

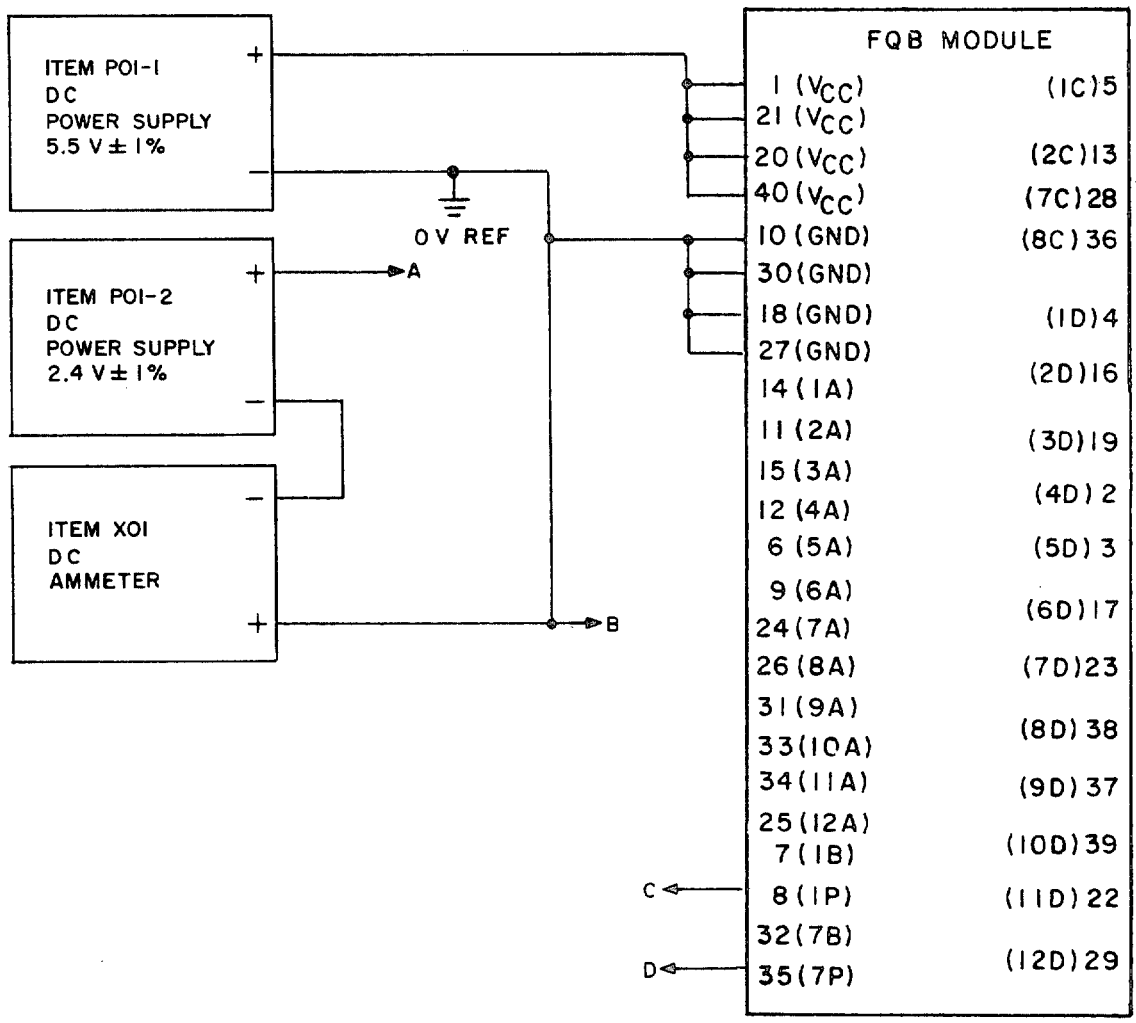
*FIGURE 17. Output current (low-level) test circuit for table XIV.

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TABLE XV. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1A Input current, high level	14	11,15,12,6,9	7	32	0	20	µA	18
P2	2A "	11	14,15,12,6,9	"	"	"	"	"	"
P3	3A "	15	14,11,12,6,9	"	"	"	"	"	"
P4	4A "	12	14,11,15,6,9	"	"	"	"	"	"
P5	5A "	6	14,11,15,12,9	"	"	"	"	"	"
P6	6A "	9	14,11,15,12,6	"	"	"	"	"	"
P7	7A "	24	26,31,33,34,25	"	"	"	"	"	"
P8	8A "	26	24,31,33,34,25	"	"	"	"	"	"
P9	9A "	31	24,26,33,34,25	"	"	"	"	"	"
P10	10A "	33	24,26,31,34,25	"	"	"	"	"	"
P11	11A "	34	24,31,33,25,26	"	"	"	"	"	"
P12	12A "	25	24,26,31,33,34	"	"	"	"	"	"
P13	1B "	7	14,11,15,12,6,9	Open	Open	"	40	"	"
P14	7B "	32	24,26,31,33,34,25	Open	Open	"	40	"	"

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CAUTION:
Do not isolate the chassis of item XO1 from earth ground.

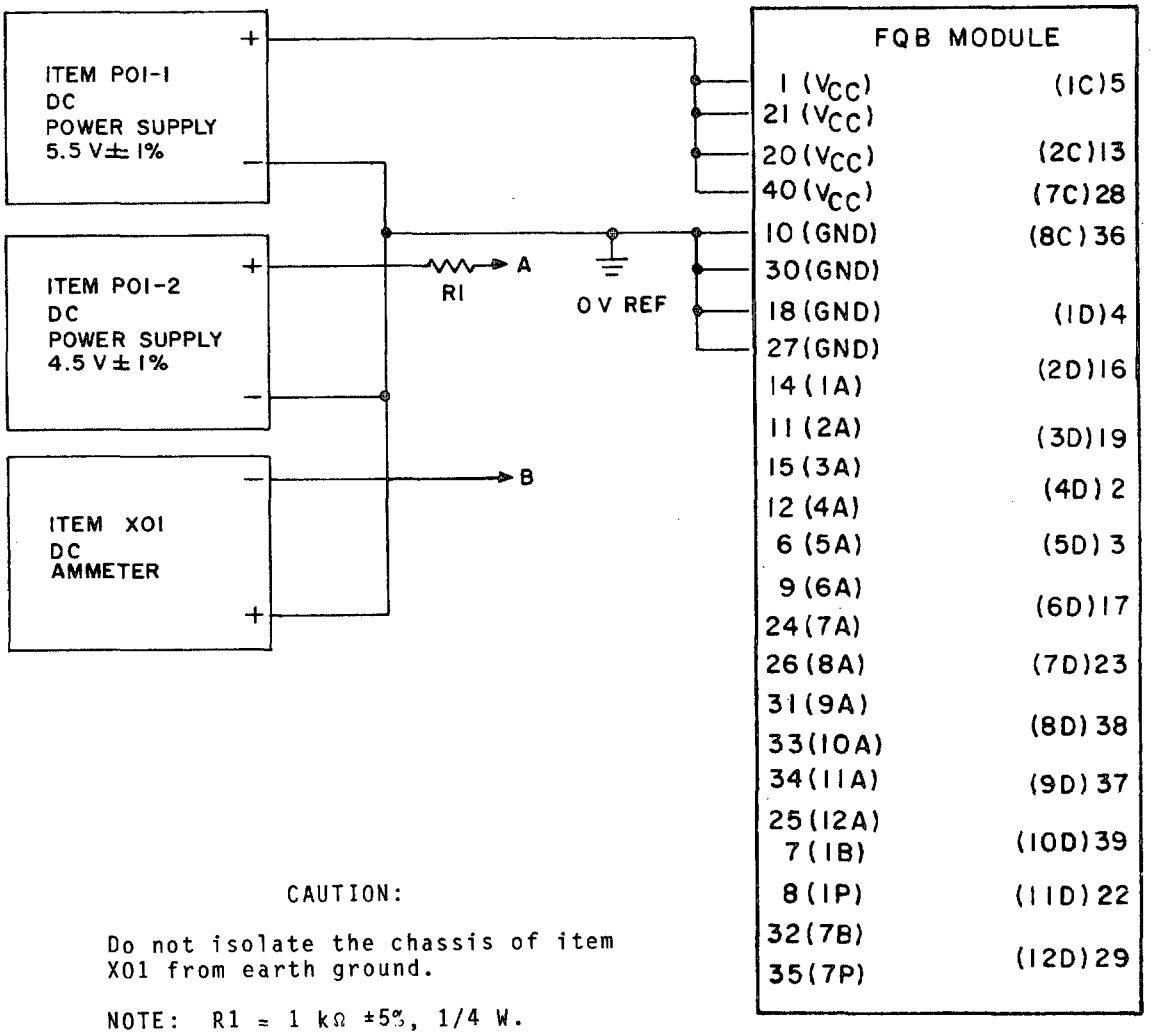
FIGURE 18. Input current (high-level) test circuit for table XV.

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TABLE XVI. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	1A Input current, low level	11,15,12,6,9,24,26,31, 33,34,25,7,32	14	-50	-400	µA	19
R2	2A "	14,15,12,6,9,24,26,31, 33,34,25,7,32	11	"	"	"	"
R3	3A "	14,11,12,6,9,24,26,31, 33,34,25,7,32	15	"	"	"	"
R4	4A "	14,11,15,6,9,24,26,31, 33,34,25,7,32	12	"	"	"	"
R5	5A "	14,11,15,12,9,24,26, 31,33,34,25,7,32	6	"	"	"	"
R6	6A "	14,11,15,12,6,24,26, 31,33,34,25,7,32	9	"	"	"	"
R7	7A "	14,11,15,12,6,9,26, 31,33,34,25,7,32	24	"	"	"	"
R8	8A "	14,11,15,12,6,9,24, 31,33,34,25,7,32	26	"	"	"	"
R9	9A "	14,11,15,12,6,9,24, 26,33,34,25,7,32	31	"	"	"	"
R10	10A "	14,11,15,12,6,9,24, 26,31,34,25,7,32	33	"	"	"	"
R11	11A "	14,11,15,12,6,9,24, 26,31,33,25,7,32	34	"	"	"	"
R12	12A "	14,11,15,12,6,9,24, 26,31,33,34,7,32	25	"	"	"	"
R13	1B "	14,11,15,12,6,9,24, 26,31,33,34,25,32	7	-0.40	-1.73	mA	"
R14	7B "	14,11,15,12,6,9,24, 26,31,33,34,25,7	32	-0.40	-1.73	mA	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: R1 = 1 kΩ ±5%, 1/4 W.

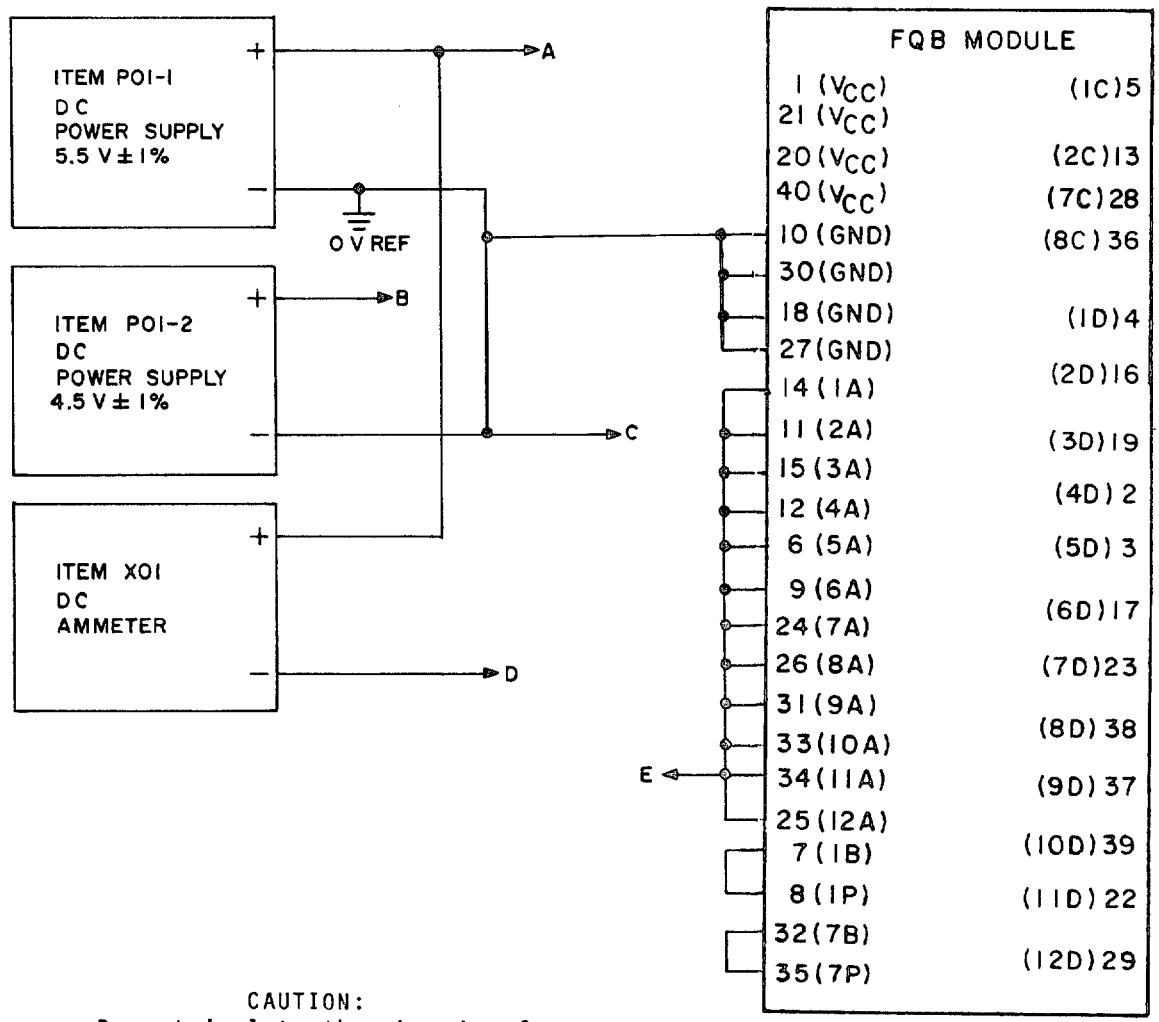
* FIGURE 19. Input current (low-level) test circuit for table XVI.

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*TABLE XVII. Power supply current (high-level and low-level outputs) tests.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Power supply current, V_{CC} logic, high level, D outputs, circuits 1-6	Open	E	Open	1	10	37	mA	20
S2	Power supply current, V_{CC} logic, high level, D outputs, circuits 7-12	"	E	Open	21	10	37	"	"
S3	Power supply current, V_{CC} logic, low level, D outputs, circuits 1-6	"	Open	E	1	15	43	"	"
S4	Power supply current, V_{CC} logic, low level, D outputs, circuits 7-12	"	Open	E	21	15	43	"	"
S5	Power supply current, V_{CC} load, high level, D outputs, circuits 1-6	1	E	Open	20	45	100	"	"
S6	Power supply current, V_{CC} load, high level, D outputs, circuits 7-12	21	E	Open	40	45	100	"	"
S7	Power supply current, V_{CC} load, low level, D outputs, circuits 1-6	1	Open	E	20	0	50	µA	"
S8	Power supply current, V_{CC} load, low level, D outputs, circuits 7-12	21	Open	E	40	0	50	µA	"

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

FIGURE 20. Power supply current (high-level and low-level outputs) test circuit for table XVII.

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	1C Transition time (t_{TLH}), C_L = 50 pF	7	5	1	35	ns	21,23
T2	7C "	32	28	"	"	"	"
T3	2C "	11	13	"	"	"	"
T4	8C "	26	36	"	"	"	"
T5	1P "	14	8	"	60	"	"
T6	7P "	24	35	"	60	"	"
T7	1C Transition time (t_{TLH}), C_L = 150 pF	7	5	"	50	"	"
T8	7C "	32	28	"	"	"	"
T9	2C "	11	13	"	"	"	"
T10	8C "	26	36	"	"	"	"
T11	1P "	14	8	"	100	"	"
T12	7P "	24	35	"	100	"	"
U1	1C Transition time (t_{THL}), C_L = 50 pF	7	5	"	15	"	"
U2	7C "	32	28	"	"	"	"
U3	2C "	11	13	"	"	"	"
U4	8C "	26	36	"	"	"	"
U5	1P "	14	8	"	30	"	"
U6	7P "	24	35	"	30	"	"
U7	1C Transition time (t_{THL}), C_L = 150 pF	7	5	"	25	"	"
U8	7C "	32	28	"	"	"	"
U9	2C "	11	13	"	"	"	"
U10	8C "	26	36	"	"	"	"

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U11	1P Transition time (t_{THL}), C_L = 150 pF	14	8	1	60	ns	21,23
U12	7P Transition time (t_{THL}), C_L = 150 pF	24	35	1	60	"	"
V1	1B/1C Propagation delay time (t_{PLH}), C_L = 50 pF	7	5	3	26	"	"
V2	7B/7C "	32	28	3	26	"	"
V3	2A/2C "	11	13	6	116	"	"
V4	8A/8C "	26	36	6	116	"	"
V5	1A/1P Propagation delay time (t_{PLH}), C_L = 50 pF	14	8	3	90	"	"
V6	7A/7P Propagation delay time (t_{PLH}), C_L = 50 pF	24	35	"	90	"	"
V7	1B/1C Propagation delay time (t_{PLH}), C_L = 150 pF	7	5	"	40	"	"
V8	7B/7C "	32	28	"	40	"	"
V9	2A/2C "	11	13	6	130	"	"
V10	8A/8C "	26	36	6	130	"	"
V11	1A/1P "	14	8	3	100	"	"
V12	7A/7P "	24	35	"	100	"	"
W1	1B/1C Propagation delay time (t_{PHL}), C_L = 50 pF	7	5	"	35	"	"
W2	7B/7C "	32	28	"	35	"	"
W3	2A/2C "	11	13	6	120	"	"
W4	8A/8C "	26	36	6	120	"	"
W5	1A/1P "	14	8	3	90	"	"
W6	7A/7P "	24	35	3	90	"	"

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

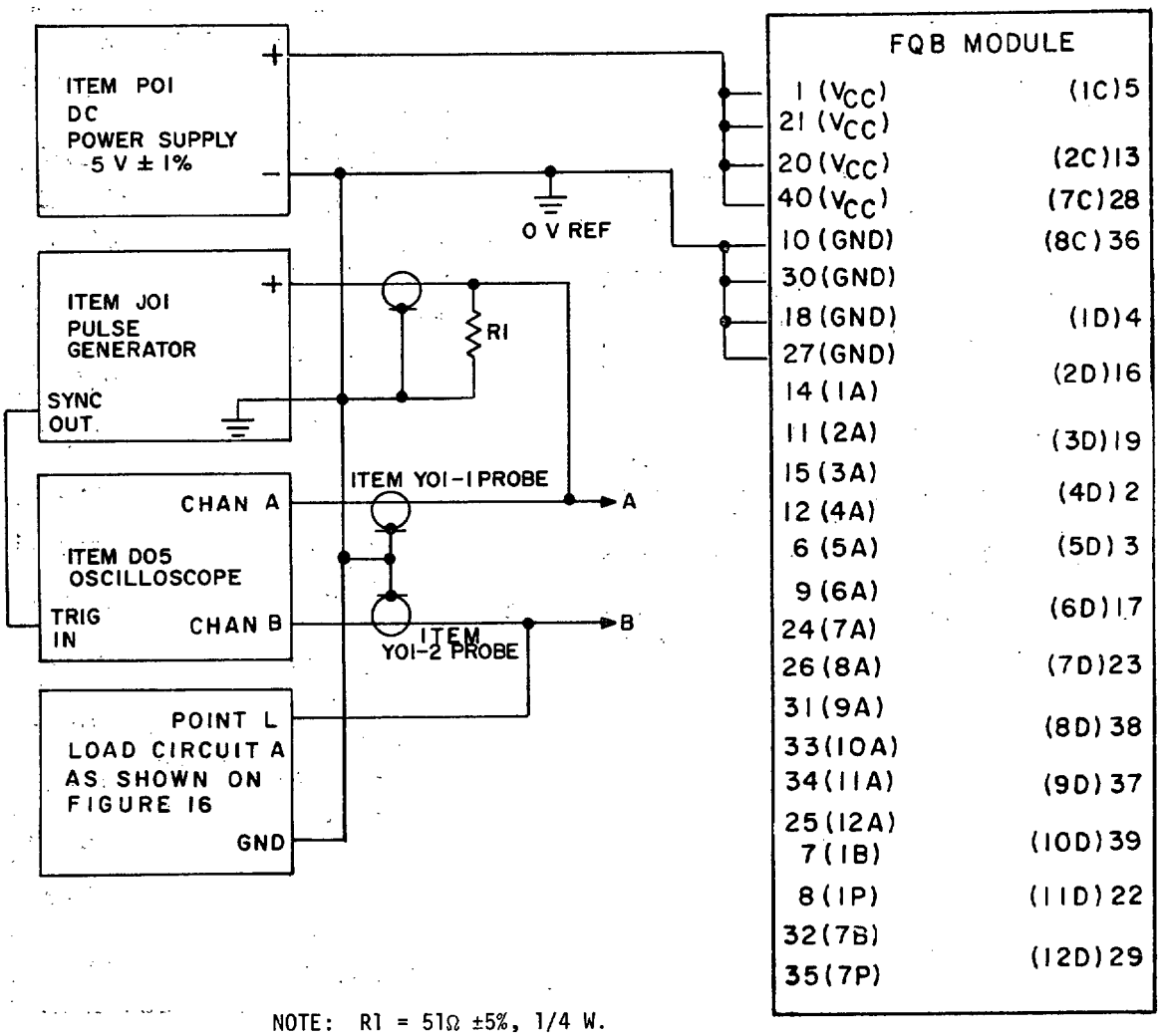
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W7	1B/1C	Propagation delay time (t_{PHL}), C_L = 150 pF	7	5	3	45	ns	21,23
W8	7B/7C	"	32	28	3	45	"	"
W9	2A/2C	"	11	13	6	150	"	"
W10	8A/8C	"	26	36	6	150	"	"
W11	1A/1P	"	14	8	3	115	"	"
W12	7A/7P	"	24	35	3	115	"	"
W13	1A/1D	Propagation delay time (t_{PLH})	14	4	1	1,000	"	22,23
W14	2A/2D	"	11	16	"	"	"	"
W15	3A/3D	"	15	19	"	"	"	"
W16	4A/4D	"	12	2	"	"	"	"
W17	5A/5D	"	6	3	"	"	"	"
W18	6A/6D	"	9	17	"	"	"	"
W19	7A/7D	"	24	23	"	"	"	"
W20	8A/8D	"	26	38	"	"	"	"
W21	9A/9D	"	31	37	"	"	"	"
W22	10A/10D	"	33	39	"	"	"	"
W23	11A/11D	"	34	22	"	"	"	"
W24	12A/12D	"	25	29	"	"	"	"
W25	1A/1D	Propagation delay time (t_{PHL})	14	4	0.01	10.00	µs	"
W26	2A/2D	"	11	16	"	"	"	"
W27	3A/3D	"	15	19	"	"	"	"
W28	4A/4D	"	12	2	"	"	"	"

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TABLE XVIII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W29	5A/5D Propagation delay time (t_{PHL})	6	3	0.01	10.00	µs	22,23
W30	6A/6D "	9	17	"	"	"	"
W31	7A/7D "	24	23	"	"	"	"
W32	8A/8D "	26	38	"	"	"	"
W33	9A/9D "	31	37	"	"	"	"
W34	10A/10D "	33	39	"	"	"	"
W35	11A/11D "	34	22	"	"	"	"
W36	12A/12D "	25	29	"	"	"	"

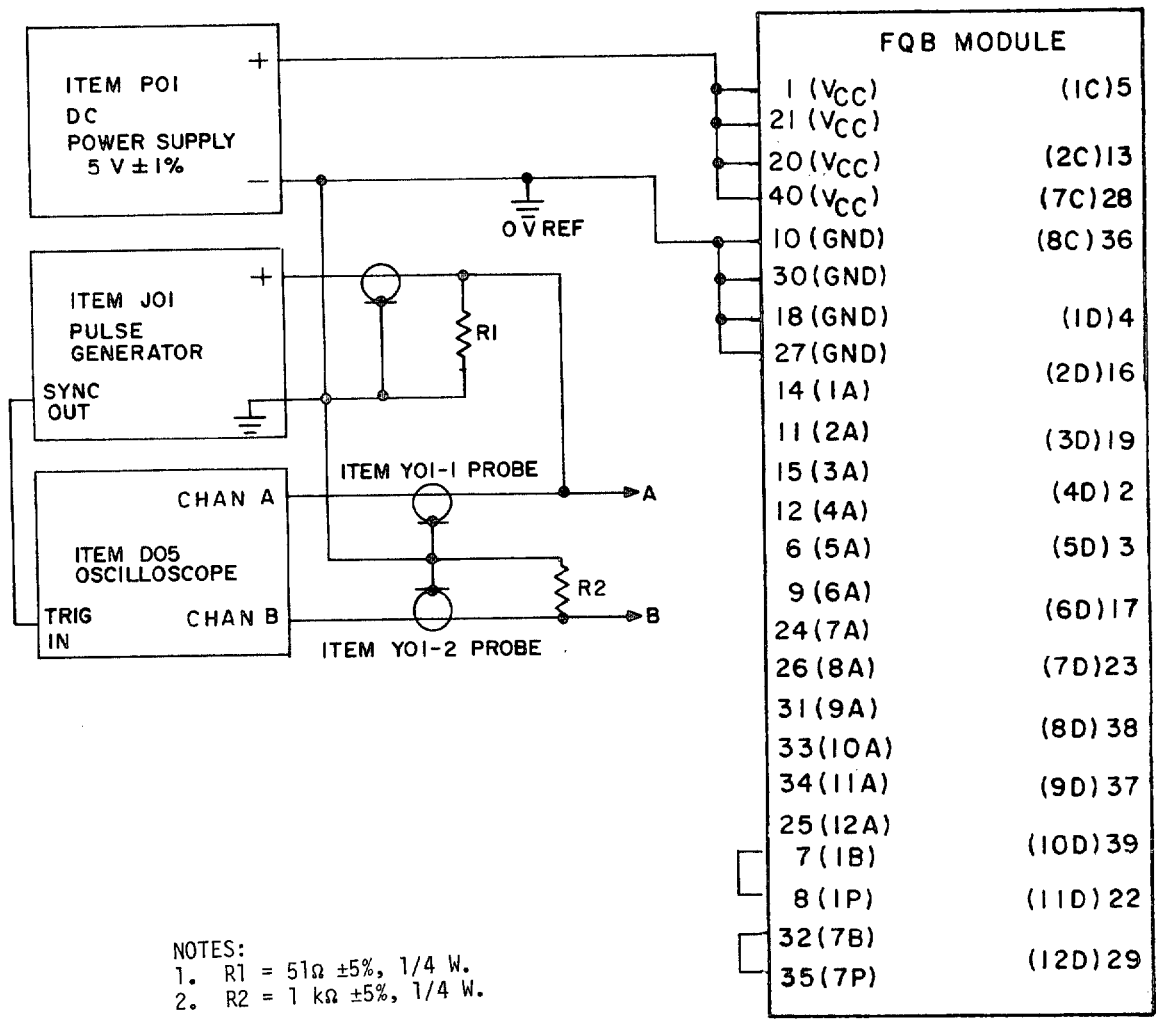
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NOTE: R1 = 51Ω ±5%, 1/4 W.

* FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVIII.

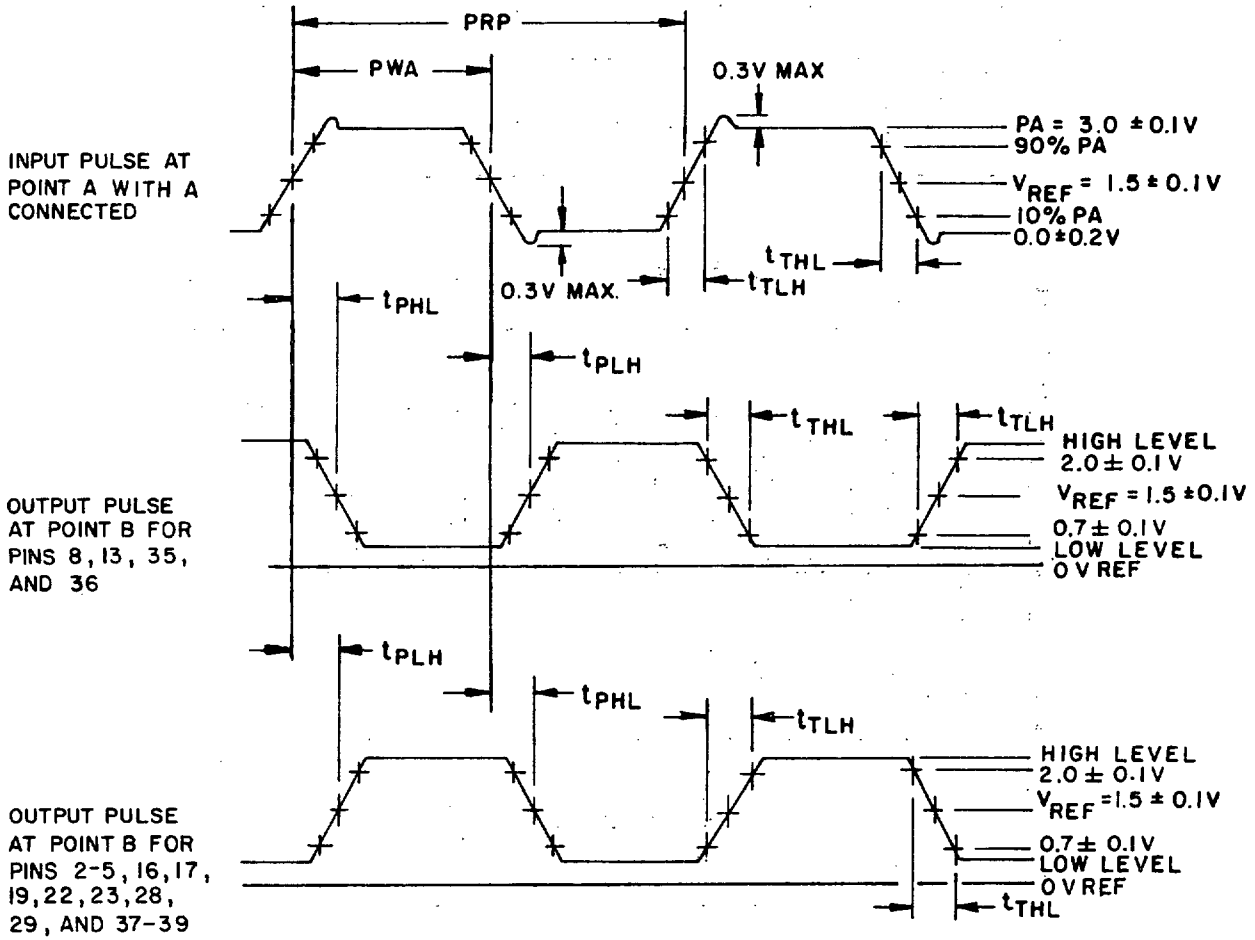
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NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. R2 = 1 kΩ ±5%, 1/4 W.

* FIGURE 22. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVIII.

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NOTES:
1. Input pulse parameters for test conditions W1-W12:
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 500 ±50 ns.
2. Input pulse parameters for test conditions W13-W36:
   PRP = 15.0 ±1.5 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 2,000 ±200 ns.
3. Before making the measurements, verify that the output pulse high level $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
4. All voltages are referenced to 0 V REF unless otherwise specified.

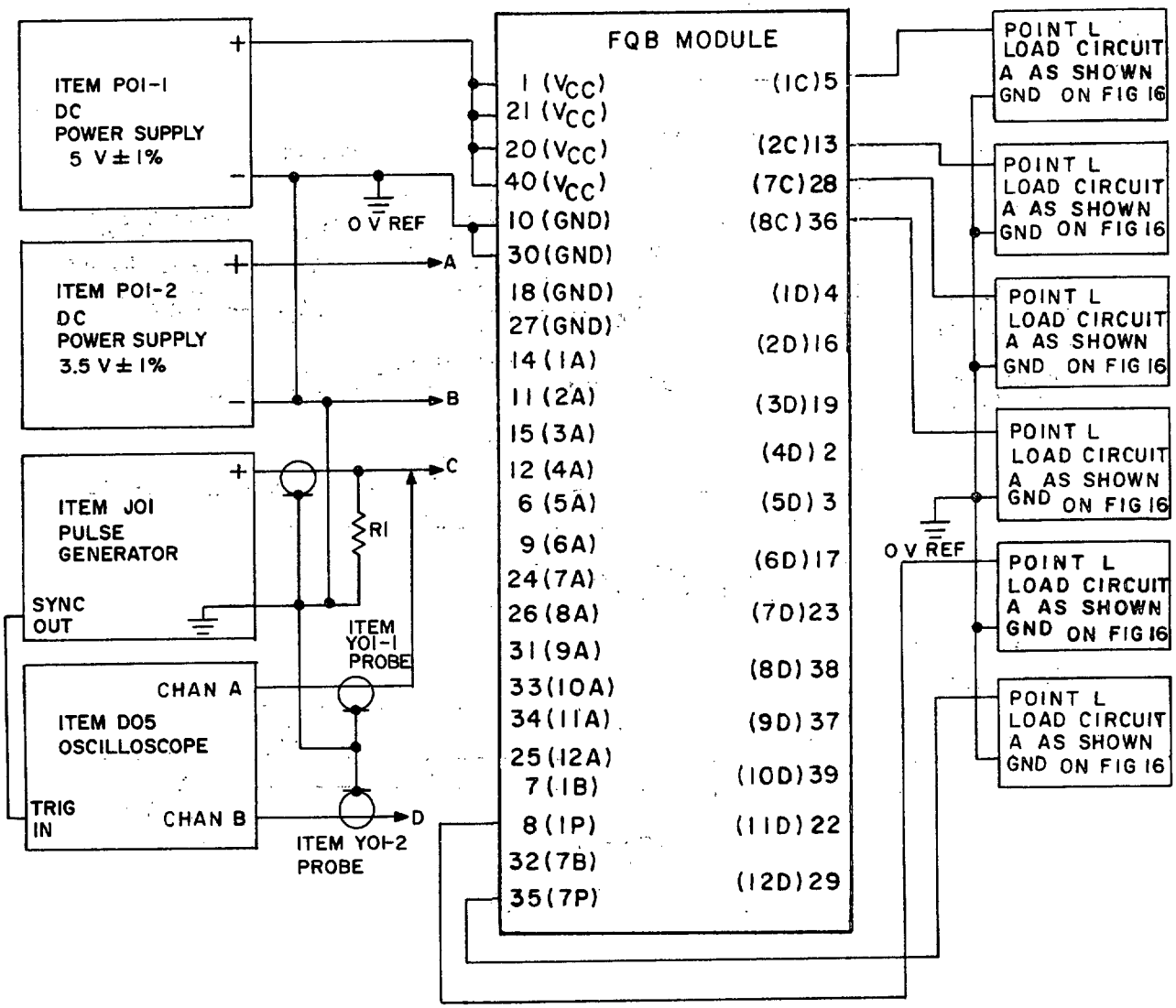
* FIGURE 23. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVIII.

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*TABLE XIX. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	1C Output crosstalk, low level	Open	7	14,11,24,32,26	5	0	400	mV	24,25
X2	2C "	11	Open	14,7,24,32,26	13	"	"	"	"
X3	7C "	Open	32	14,7,24,11,26	28	"	"	"	"
X4	8C "	26	Open	14,7,11,24,32	36	"	"	"	"
X5	1P "	14	"	7,11,24,32,26	8	"	"	"	"
X6	7P "	24	"	14,7,11,32,26	35	"	"	"	"
Y1	1C Output crosstalk, high level	7	"	14,11,24,32,26	5	"	"	"	"
Y2	2C "	Open	11	14,7,24,32,26	13	"	"	"	"
Y3	7C "	32	Open	14,7,11,24,26	28	"	"	"	"
Y4	8C "	Open	26	14,7,11,24,32	36	"	"	"	"
Y5	1P "	"	14	7,11,24,32,26	8	"	"	"	"
Y6	7P "	"	24	14,7,11,32,26	35	"	"	"	"

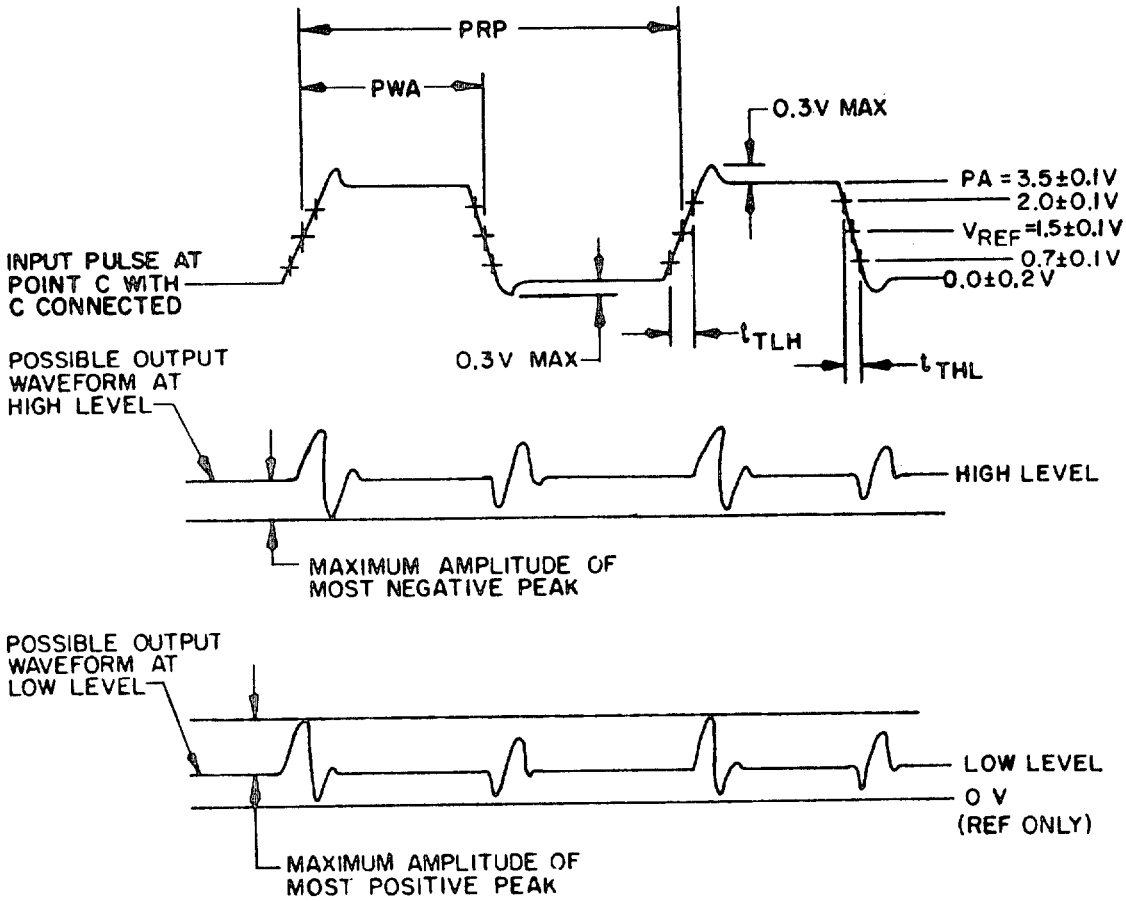
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NOTE: R1 = 51Ω ±5%, 1/4 W.

* FIGURE 24. Output crosstalk tests (low-level and high-level) test circuit for table XIX.

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NOTES:
1. Input pulse parameters:
   PRP = 20 ±2 µs.
   $t_{THL} = t_{TLH}$ = 3.5 ±1.5 ns (verified at each input).
   PWA = 10 ±1 µs.
2. All voltages are referenced to 0 V unless otherwise specified.

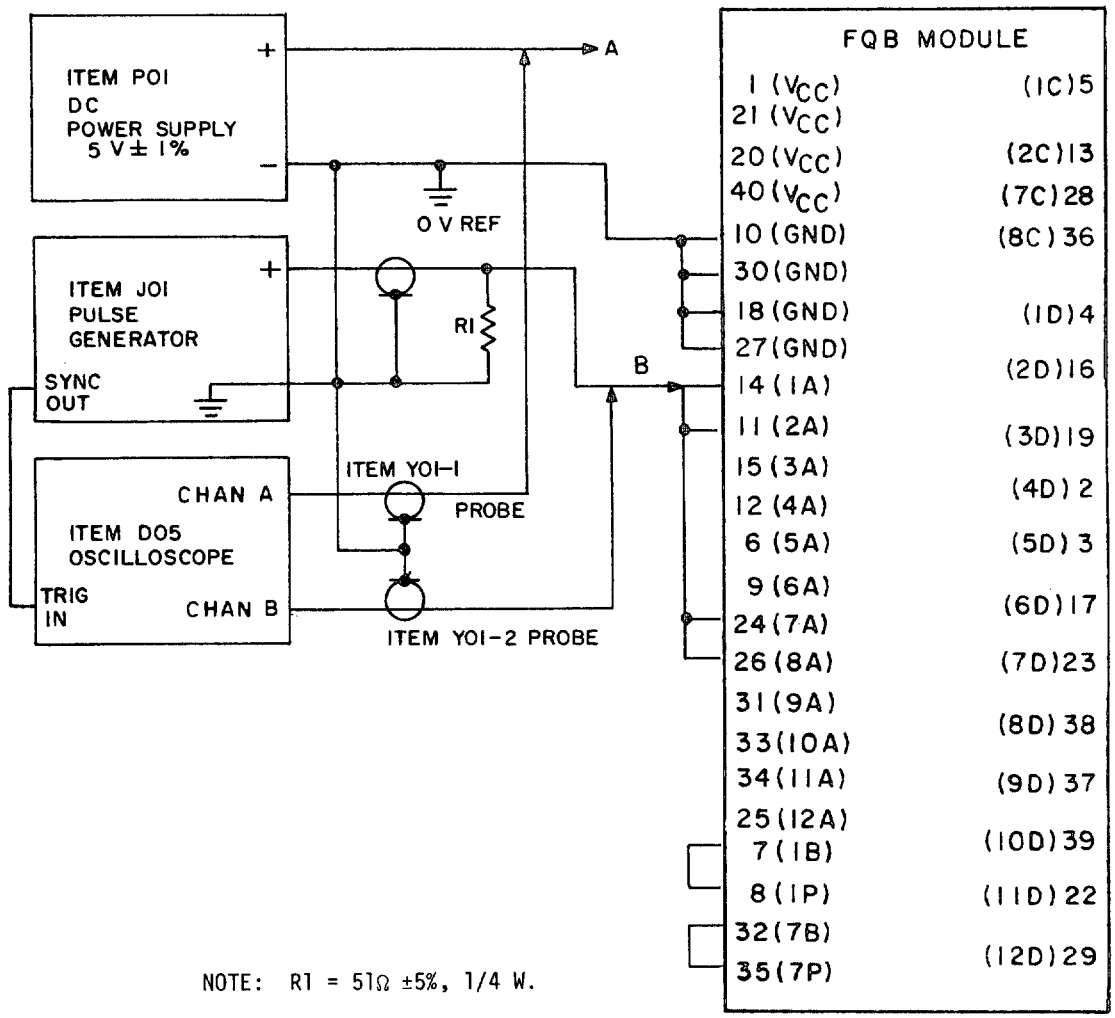
FIGURE 25. Output crosstalk (low-level and high-level) test circuit waveforms for table XIX.

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*TABLE XX. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Z1	Power supply transient amplitude (circuits 1-6)	1,20	0	500	mV	26,27
Z2	Power supply transient amplitude (circuits 7-12)	21,40	0	500	mV	26,27

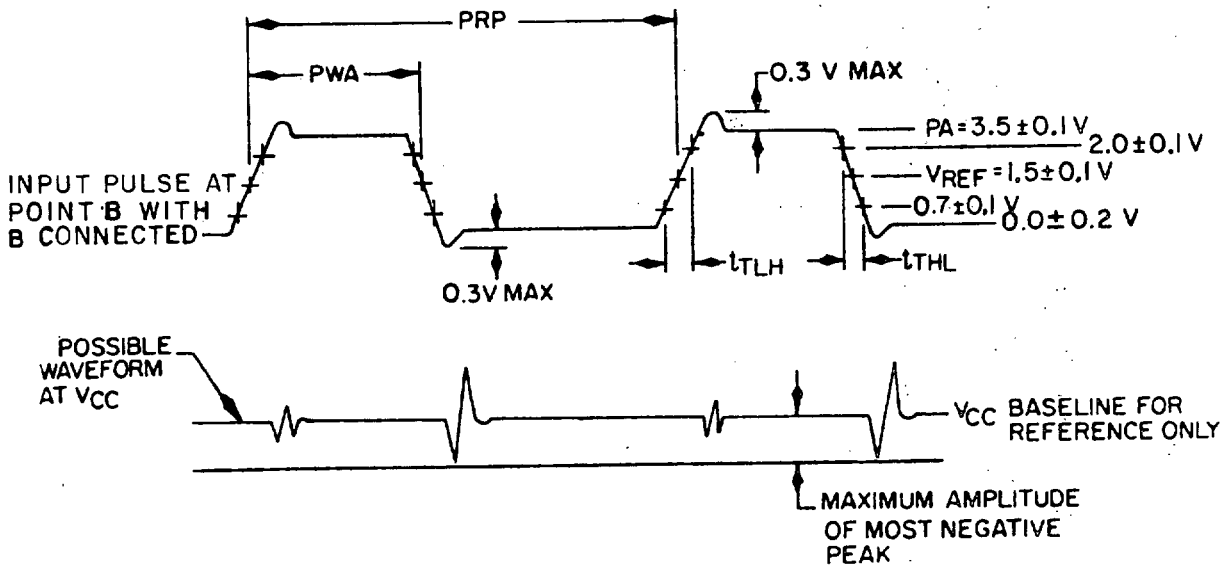
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NOTE: R1 = 51Ω ±5%, 1/4 W.

FIGURE 26. Power supply transient amplitude test circuit for table XX.

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NOTES:
1. Input pulse parameters:
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input).
   PWA = 500 ±50 ns.
2. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 27. Power supply transient amplitude test circuit waveforms for table XX.

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TABLE XXI. Power supply filter capacitance test.

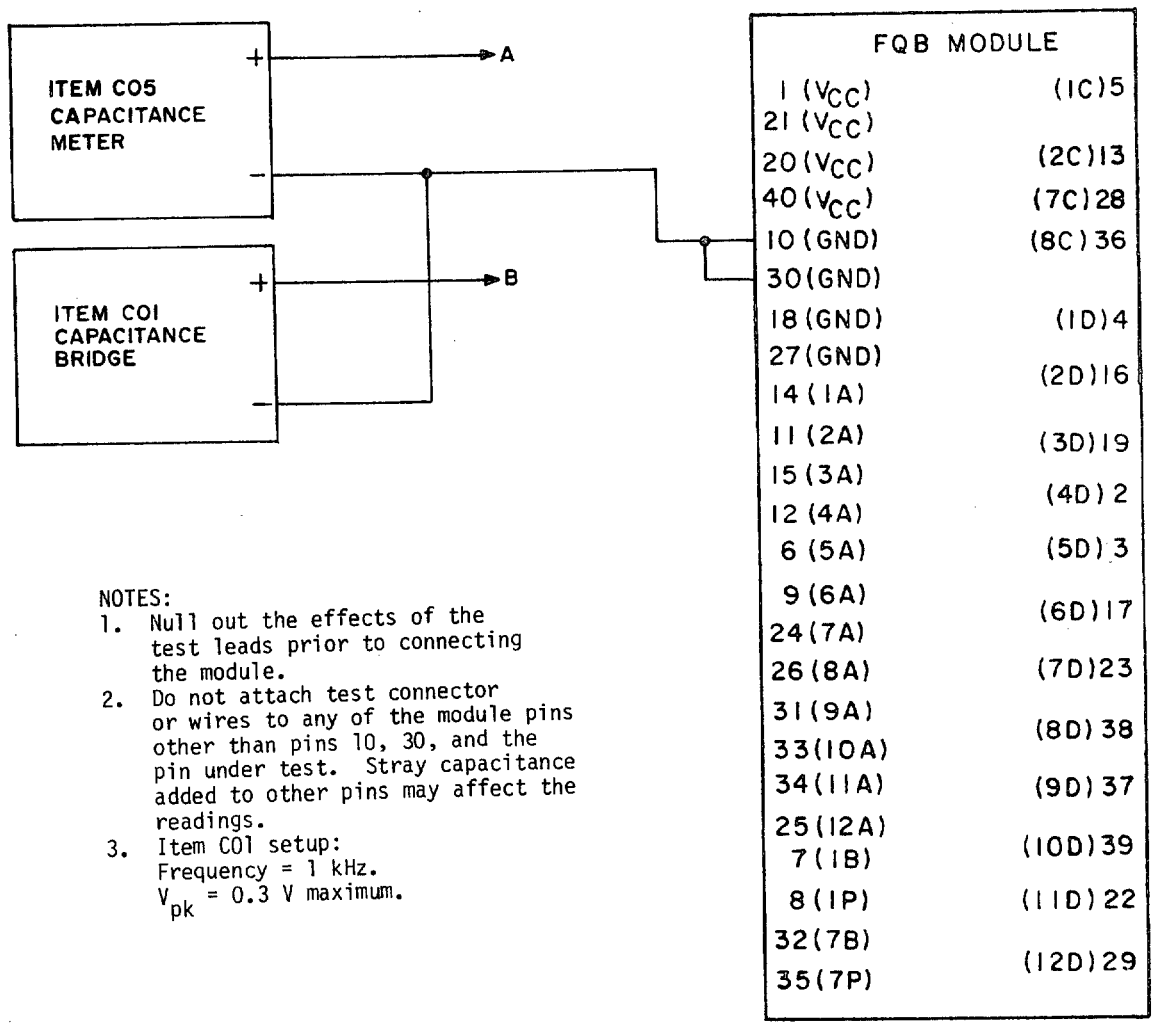
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	Power supply filter capacitance (circuits 1-6)	Open	1	0.20	0.34	µF	28
AA2	Power supply filter capacitance (circuits 7-12)	Open	21	0.20	0.34	µF	28

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TABLE XXII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	1A Input capacitance	14	Open	2	15	pF	28
BB2	2A "	11	"	"	"	"	"
BB3	3A "	15	"	"	"	"	"
BB4	4A "	12	"	"	"	"	"
BB5	5A "	6	"	"	"	"	"
BB6	6A "	9	"	"	"	"	"
BB7	7A "	24	"	"	"	"	"
BB8	8A "	26	"	"	"	"	"
BB9	9A "	31	"	"	"	"	"
BB10	10A "	33	"	"	"	"	"
BB11	11A "	34	"	"	"	"	"
BB12	12A "	25	"	"	"	"	"
BB13	1B "	7	"	"	"	"	"
BB14	7B "	32	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 30, and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

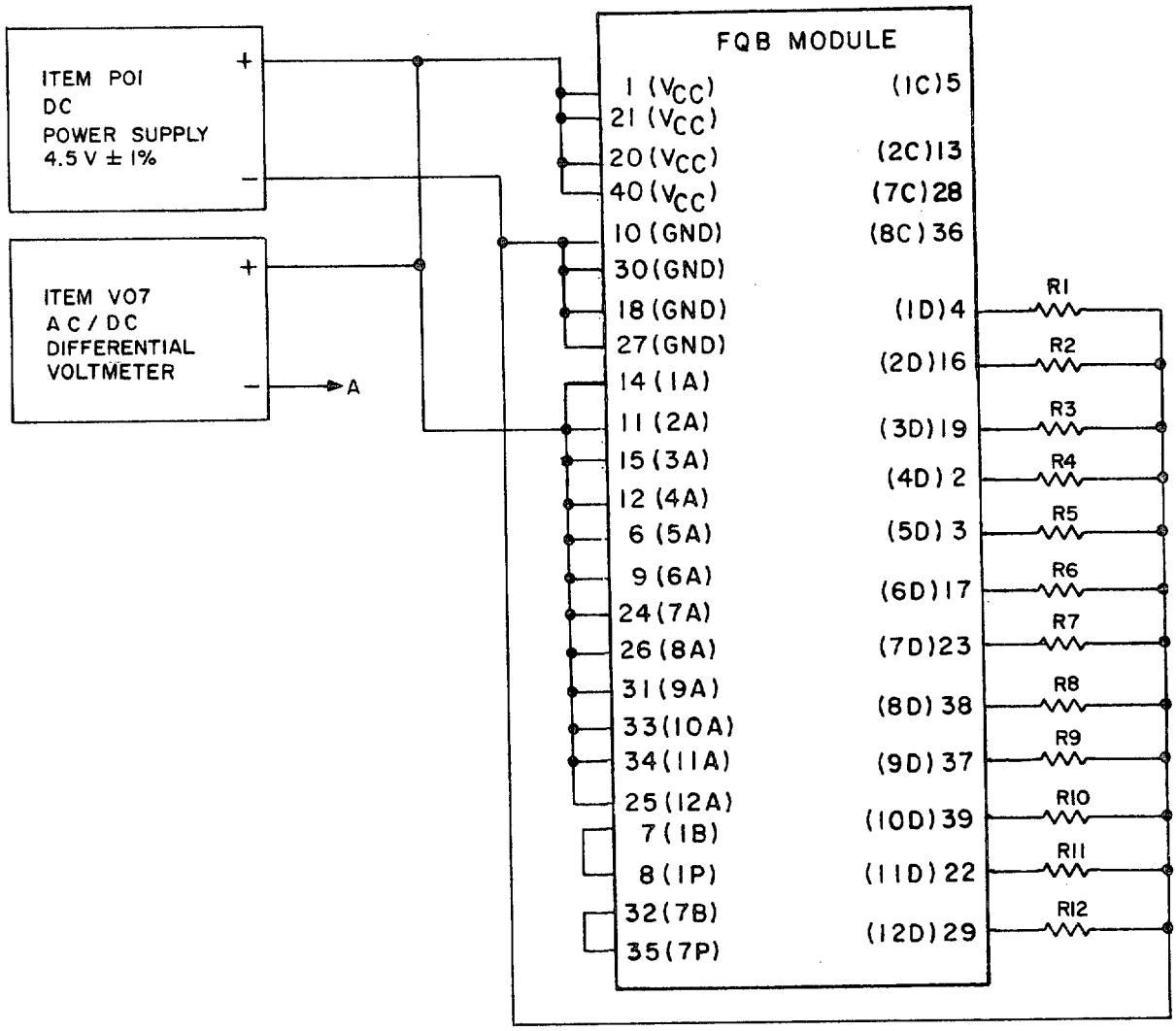
*FIGURE 28. Power supply filter and input capacitance test circuit for tables XXI and XXII.

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*TABLE XXIII. Saturation voltage (full load) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
CC1	1D Saturation voltage, full load	4	10	350	mV	29
CC2	2D "	16	"	"	"	"
CC3	3D "	19	"	"	"	"
CC4	4D "	2	"	"	"	"
CC5	5D "	3	"	"	"	"
CC6	6D "	17	"	"	"	"
CC7	7D "	23	"	"	"	"
CC8	8D "	38	"	"	"	"
CC9	9D "	37	"	"	"	"
CC10	10D "	39	"	"	"	"
CC11	11D "	22	"	"	"	"
CC12	12D "	29	"	"	"	"

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NOTE: R1-R12 = 23.7Ω ±1%, 3 W.
(Test verifies IR drop in lead of PCB construction with full load).

*FIGURE 29. Saturation voltage (full load) test circuit for table XXIII.

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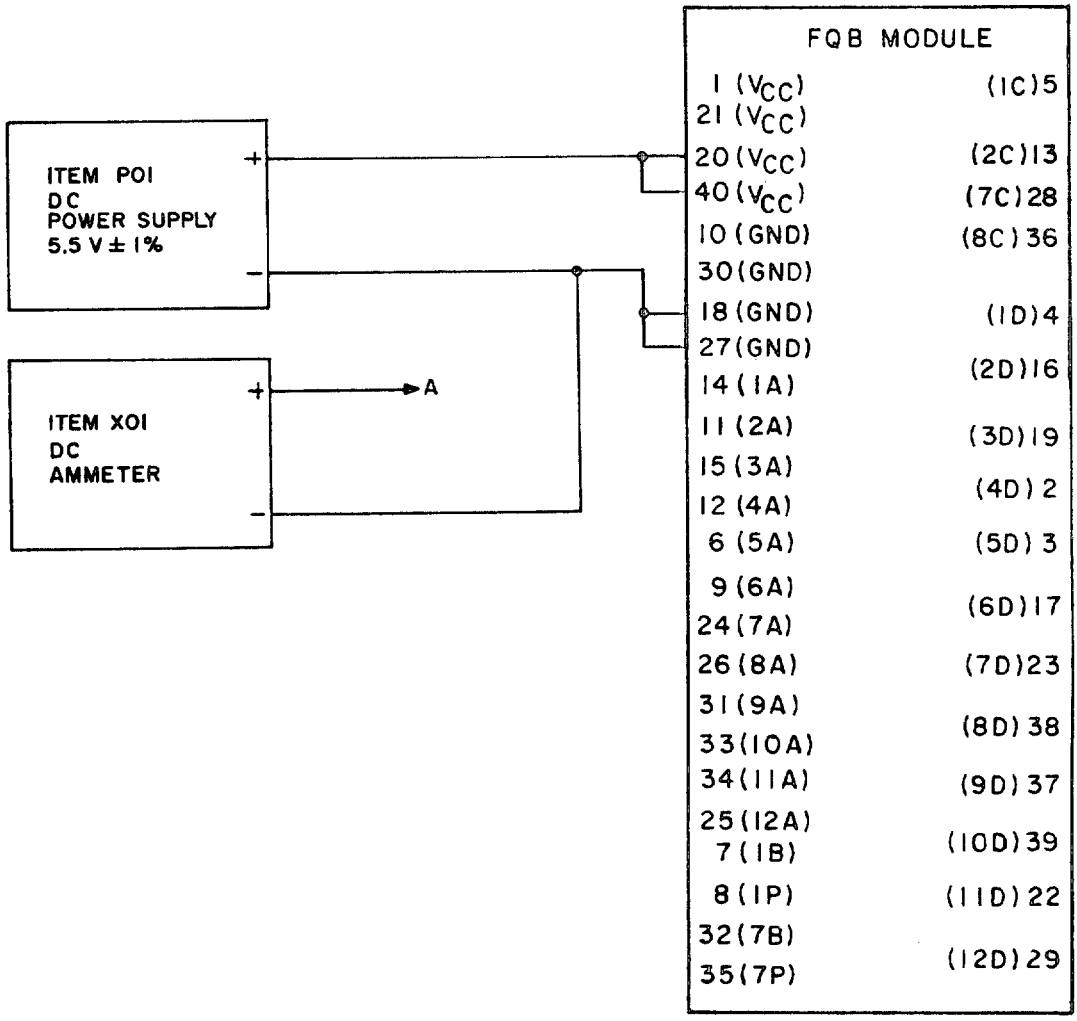
*TABLE XXIV. Low level input current (I_{IL}) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
DD1	1C Low level input current (I_{IL})	5	8	17	mA	30
DD2	2C "	13	"	"	"	"
DD3	7C "	28	"	"	"	"
DD4	8C "	36	"	"	"	"

*TABLE XXV. High level output current (I_{OH}) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
EE1	1C High level output current (I_{OH})	4	0	15	µA	31
EE2	2C "	16	"	"	"	"
EE3	7C "	23	"	"	"	"
EE4	8C "	38	"	"	"	"

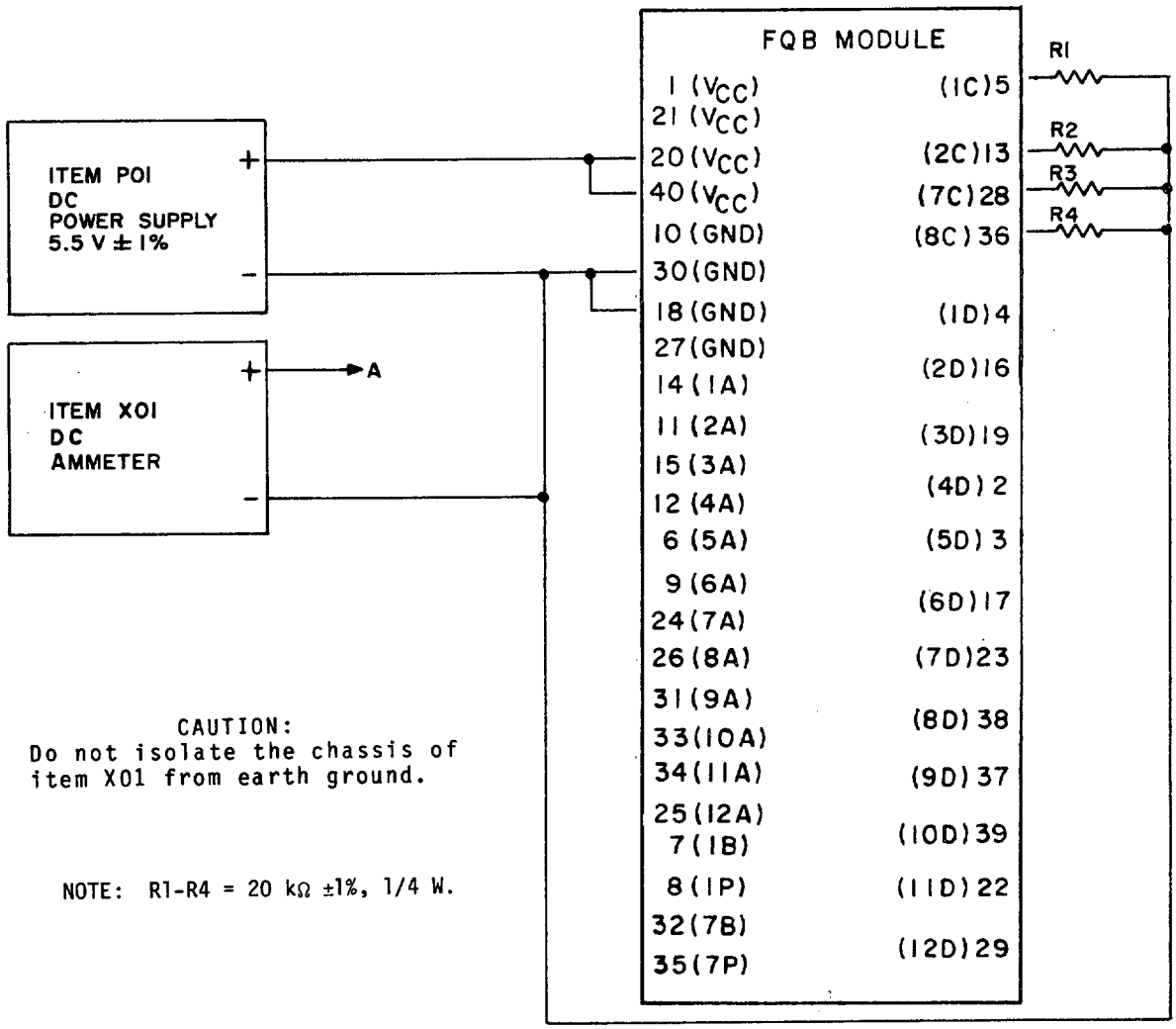
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CAUTION:
Do not isolate the chassis of
item X01 from earth ground.

*FIGURE 30. Low level input current (I_{IL}) test circuit for table XXIV.

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTE: R1-R4 = 20 kΩ ±1%, 1/4 W.

*FIGURE 31. High level output current (I_{OH}) test circuit for table XXV.

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